Image-Identifiable Genomic Subspaces: A Linear–Gaussian Model of Radiogenomic Recoverability

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Abstract

We develop a linear–Gaussian generative model linking patient-level genomic profiles $g \in \mathbb{R}^p$ to imaging phenotypes $x \in \mathbb{R}^d$, and use it to ask a sharper question than "does imaging predict gene j?" Instead we ask *which low-dimensional genomic subspaces are image-identifiable*. Under the model every quantity of interest is available in closed form: the Bayes-optimal estimator $\mathbb{E}[g \mid x]$, the posterior covariance $\Sigma_{G|X}$, the mutual information $I(g; x)$, and a per-direction *recoverability score* $R(v) \in [0, 1]$. We show that the directions of maximal recoverability solve a generalized eigenproblem that coincides exactly with canonical correlation analysis (CCA) between the two modalities, and that the recoverability of the i-th canonical genomic direction equals the squared canonical correlation ρ_i^2. We then address estimation in the regime $p \gg n$ via regularized covariances, reduced-rank regression, and the probabilistic-CCA latent-variable form, and describe cross-validated and permutation-based procedures for honest inference, including a scale-matched permutation null that prevents high-dimensional inflation from masquerading as signal. The mathematics is deliberately classical; the contributions are the recoverability framing, the honest-rank estimation pipeline, and a cautionary three-cohort study. Applied to TCGA-KIRC, an NSCLC cohort, and ADNI, the same pipeline returns three different verdicts: only TCGA-KIRC shows a small ($\hat{R}^{\text{cv}} \approx 0.1$, single-direction) association that survives both cross-validation and demographic/scanner deconfounding, whereas the apparent NSCLC signal is cohort-specific overfit and the ADNI signal an age/sex artefact — all three of which a naive in-sample estimate would report as strong positives. The companion notebook `notebooks/radiogenomic_recoverability.ipynb` implements the full pipeline on a pair of `AnnData` objects, with a synthetic-data fallback that has known ground truth.

Keywords radiogenomics · canonical correlation analysis · mutual information · recoverability · linear–Gaussian model

1 Introduction

Radiogenomics asks how much of a tumour's — or an organ's — molecular state is written into its image [1–4]. The dominant empirical paradigm trains one predictor per gene or per signature and reports where imaging "predicts" genomics. At the cohort sizes typical of the field ($n \sim 10^2$) against panels of 10^3–10^4 transcripts, this is a high-dimensional, low-power multiple-testing problem in which apparent positives are easily manufactured by overfitting, batch structure, or demographic confounding. We argue that the more useful and more answerable question is not "does imaging predict gene j?" but *how many genomic directions are image-identifiable at all, and how well* — a low-dimensional, estimable quantity with a hard ceiling set by the biology–imaging channel rather than by the size of the gene panel.

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To make this precise we adopt the simplest model with a non-trivial answer: a linear–Gaussian channel from genomics $g \in \mathbb{R}^p$ to imaging $x \in \mathbb{R}^d$. Under it every quantity of interest is closed-form — the Bayes-optimal estimator $\mathbb{E}[g \mid x]$, the posterior covariance, the mutual information $I(g; x)$, and a per-direction *recoverability score* $R(v) \in [0, 1]$ — and the directions of maximal recoverability coincide exactly with canonical correlation analysis (CCA) between the two modalities, with $R(a_i) = \rho_i^2$, the squared canonical correlation. The model thus turns a vague "fundamental limit" intuition into a concrete, estimable spectrum $\rho_1^2 \geq \rho_2^2 \geq \cdots$ and an associated set of genomic directions that can be read back onto genes and pathways.

2 Related work

CCA and its probabilistic form. Canonical correlation analysis dates to Hotelling [5]; its probabilistic / latent-variable interpretation, which we use in Section 12.4, is due to Bach and Jordan [6], and the kernel/regularized variants are surveyed by Hardoon et al. [7]. The identity that links Gaussian mutual information to the canonical correlations, $I = -\frac{1}{2}\sum_i \log(1 - \rho_i^2)$, is classical [8]. Our recoverability score and eigenproblem are a re-reading of this machinery in terms of posterior-variance reduction per genomic direction; we add the rank limit and the binary-data extension.

High-dimensional and sparse CCA. When p, d are comparable to n, sample canonical correlations are severely biased upward. The limiting laws of sample canonical correlations under the null and finite-rank alternatives are characterized by random-matrix theory [9–11], which we use directly to place a detection edge on the spectrum (Section 12.5). Two families of remedies exist: regularization/shrinkage (Ledoit–Wolf [12]) and sparsity (penalized CCA, 13; sparse-PCA consistency theory, 14). We adopt symmetric dimension reduction plus shrinkage and lean on cross-validation and permutation for honesty rather than on sparsity, but sparse CCA is a natural alternative estimator within the same framing.

Information-theoretic limits and estimation. The fundamental-limit perspective — bounding what *any* predictor can achieve — is in the spirit of rate–distortion theory [15] and the information bottleneck [16–18], specialized here to a linear–Gaussian channel where the bottleneck is exactly solvable. Because we report mutual information, we are mindful that general MI estimation is fragile in high dimensions and that variational estimators can be badly biased [19–21]; our reported MI is the closed-form Gaussian functional of estimated canonical correlations, not a generic estimator, and we bound its misspecification error in both directions (Sections 12.7, 15.4).

Radiogenomics. Empirical radiogenomics has reported imaging correlates of molecular state across many tumour types [22–28], typically as per-gene or per-signature predictors built on radiomic features [29–31] or deep embeddings [32]. Our framework is complementary: it does not build a predictor but bounds and counts the directions any predictor could exploit, and it supplies the confound and sample-size controls that distinguish a real low-rank channel from the overfitting and batch effects that the per-gene paradigm is prone to.

3 Setup and notation

We observe n patients. For patient i let $g_i \in \mathbb{R}^p$ be a vector of genomic features (e.g. expression, somatic mutation burden, copy-number scores) and $x_i \in \mathbb{R}^d$ a vector of imaging phenotypes (radiomic descriptors, deep features, or structured reads) [2, 29, 33]. Stack the observations into data matrices $\mathbf{G} \in \mathbb{R}^{n \times p}$ and $\mathbf{X} \in \mathbb{R}^{n \times d}$, with row i equal to $g_i^\top$ and $x_i^\top$ respectively. Random variables are written in uppercase (G, X), realizations in lowercase (g, x). Throughout we assume both modalities have been centered, so all means are zero; recentring is handled by the estimators of Section 12.

We use $\Sigma_G \in \mathbb{R}^{p \times p}$, $\Sigma_X \in \mathbb{R}^{d \times d}$ for the marginal covariances, $\Sigma_{GX} = \mathrm{Cov}(G, X) \in \mathbb{R}^{p \times d}$ for the cross-covariance, and $\Sigma_{XG} = \Sigma_{GX}^\top$. For a symmetric positive semidefinite S we write $S^{1/2}$ for its symmetric square root and $S^{-1/2}$ for the inverse thereof.

4 The linear–Gaussian generative model

The genomic state is drawn from a Gaussian prior and the imaging phenotype is a noisy linear readout of it:

$$G \sim \mathcal{N}(0, \Sigma_G), \tag{1}$$

$$X \mid G \sim \mathcal{N}(AG,\ \sigma^2 I_d), \qquad A \in \mathbb{R}^{d \times p}. \tag{2}$$

The matrix A encodes how genomic variation is expressed in the image; the isotropic term $\sigma^2 I_d$ collects everything in the image that genomics does not drive — biological variation outside the measured panel, acquisition and reconstruction noise, and feature-extraction error. Equivalently,

$$X = AG + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \sigma^2 I_d), \quad \varepsilon \perp G. \tag{3}$$

Equations (1)–(3) are deliberately the simplest model with a non-trivial answer. Their value is that *every* downstream quantity is closed-form, so the model functions as an exact upper bound: no estimator — linear or nonlinear — can recover genomics from imaging better than the posterior derived below (under these assumptions). This fundamental-limit framing is in the spirit of rate–distortion theory and the information bottleneck [15–18, 34], specialized here to a tractable linear–Gaussian channel. Section 12 relaxes the isotropic-noise and known-parameter assumptions for practical fitting.

5 Marginal and joint laws

Proposition 1 (Joint Gaussian law; classical Gaussian identity, e.g. 8)**.** *Under* (1)–(3)*,*

$$\begin{pmatrix} G \\ X \end{pmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{pmatrix} \Sigma_G & \Sigma_G A^\top \\ A\Sigma_G & A\Sigma_G A^\top + \sigma^2 I_d \end{pmatrix}\right). \tag{4}$$

In particular,

$$X \sim \mathcal{N}(0, \Sigma_X), \qquad \Sigma_X = A\Sigma_G A^\top + \sigma^2 I_d, \tag{5}$$

and the cross-covariance is

$$\Sigma_{GX} = \mathrm{Cov}(G, X) = \Sigma_G A^\top. \tag{6}$$

Proof. Since G and ε are independent Gaussians and (G, X) is the affine image $(G,\, AG + \varepsilon)$, the pair is jointly Gaussian with zero mean. The covariance blocks follow by standard Gaussian conditioning: $\mathrm{Cov}(G, G) = \Sigma_G$, $\mathrm{Cov}(G, X) = \mathrm{Cov}(G, AG + \varepsilon) = \Sigma_G A^\top$ (using $\varepsilon \perp G$), and $\mathrm{Var}(X) = \mathrm{Cov}(AG + \varepsilon) = A\Sigma_G A^\top + \sigma^2 I_d$. □

Equation (5) is the model's first interpretive statement: imaging covariance decomposes additively into a *genomics-driven* component $A\Sigma_G A^\top$ and an *irreducible* component $\sigma^2 I_d$. The cross-covariance $\Sigma_{GX} = \Sigma_G A^\top$ is the central estimable object — it names which genomic directions co-vary with imaging at all.

6 Bayes-optimal recovery of genomics from imaging

Proposition 2 (Posterior; standard Gaussian conditioning)**.** *The posterior of genomics given imaging is Gaussian,* $G \mid X \sim \mathcal{N}(\hat{g}(X), \Sigma_{G|X})$*, with*

$$\hat{g}(X) = \mathbb{E}[G \mid X] = \Sigma_G A^\top \big(A\Sigma_G A^\top + \sigma^2 I_d\big)^{-1} X \;=\; BX, \tag{7}$$

$$\Sigma_{G|X} = \Sigma_G - \Sigma_G A^\top \big(A\Sigma_G A^\top + \sigma^2 I_d\big)^{-1} A\Sigma_G. \tag{8}$$

Equivalently, in precision form,

$$\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A, \qquad B = \sigma^{-2}\, \Sigma_{G|X}\, A^\top. \tag{9}$$

Proof. By standard Gaussian conditioning on the joint law of Proposition 1, $G \mid X$ is Gaussian with mean $\Sigma_{GX}\Sigma_X^{-1} X$ and covariance $\Sigma_G - \Sigma_{GX}\Sigma_X^{-1}\Sigma_{XG}$. Substituting the blocks $\Sigma_{GX} = \Sigma_G A^\top$, $\Sigma_{XG} = A\Sigma_G$, $\Sigma_X = A\Sigma_G A^\top + \sigma^2 I_d$ gives (7)–(8) and $B = \Sigma_G A^\top \Sigma_X^{-1}$. The precision form $\Sigma_{G|X}^{-1} = \Sigma_G^{-1} + \sigma^{-2} A^\top A$ and $B = \sigma^{-2}\Sigma_{G|X} A^\top$ follow from the Woodbury identity applied to $\Sigma_{G|X}$ and the push-through identity $A^\top(A\Sigma_G A^\top + \sigma^2 I_d) = \sigma^2(\Sigma_G^{-1} + \sigma^{-2} A^\top A)\Sigma_G A^\top$. □

The estimator $\hat{g}(X) = BX$ is the Bayes-optimal predictor under squared-error loss; because the model is jointly Gaussian it is also the optimal predictor over *all* measurable functions of X, not merely linear ones. Thus, *under the assumed linear–Gaussian law*, $\Sigma_{G|X}$ is a hard floor on residual genomic uncertainty: **no algorithm can do better**. This is a statement about the model, not an unconditional guarantee on real data; Section 12.7 quantifies how far each cohort departs from the law and what that costs the bound. The precision form (9) makes the accounting transparent — imaging *adds* information $\sigma^{-2} A^\top A$ to the prior precision Σ_G^{-1}.

7 Mutual information and the SNR decomposition

Proposition 3 (Mutual information; classical Gaussian MI identity, 8)**.** *The mutual information between the two modalities is*

$$I(G;X) = \tfrac{1}{2}\log\frac{\det\Sigma_G}{\det\Sigma_{G|X}} = \tfrac{1}{2}\log\det\bigl(I_d + \sigma^{-2}A\Sigma_G A^\top\bigr). \tag{10}$$

Let $\lambda_1 \geq \cdots \geq \lambda_r > 0$ be the nonzero eigenvalues of $A\Sigma_G A^\top$. Then

$$I(G;X) = \tfrac{1}{2}\sum_{i=1}^{r}\log\left(1+\frac{\lambda_i}{\sigma^2}\right). \tag{11}$$

Proof. By standard Gaussian conditioning the mutual information of a jointly Gaussian pair is $I(G;X) = \frac{1}{2}\log(\det\Sigma_G/\det\Sigma_{G|X})$ [8]. Taking determinants in the precision form (9) and factoring out Σ_G^{-1} gives $\det\Sigma_G/\det\Sigma_{G|X} = \det(I_p+\sigma^{-2}\Sigma_G^{1/2}A^\top A\,\Sigma_G^{1/2})$; Sylvester's identity $\det(I_p+PQ) = \det(I_d+QP)$ converts this to $\det(I_d+\sigma^{-2}A\Sigma_G A^\top)$, which is (10). Diagonalizing the symmetric PSD matrix $A\Sigma_G A^\top$ with nonzero eigenvalues λ_i, the determinant is $\prod_i(1+\lambda_i/\sigma^2)$, giving (11). □

Equation (11) is the cleanest reading of the model: each image-visible genomic direction contributes $\frac{1}{2}\log(1+\mathrm{SNR}_i)$ bits (nats), where $\mathrm{SNR}_i = \lambda_i/\sigma^2$ is the signal-to-noise ratio of that direction. Directions whose genomic signal is small relative to imaging noise contribute essentially nothing, regardless of how many genes load on them. Figure 1 illustrates the decomposition (11) on the synthetic instance introduced below: the per-direction bit contribution falls off rapidly after the first handful of canonical components, and the cumulative total saturates at $I(G;X) \approx 6.45$ bits out of the analytic upper bound implied by the model.

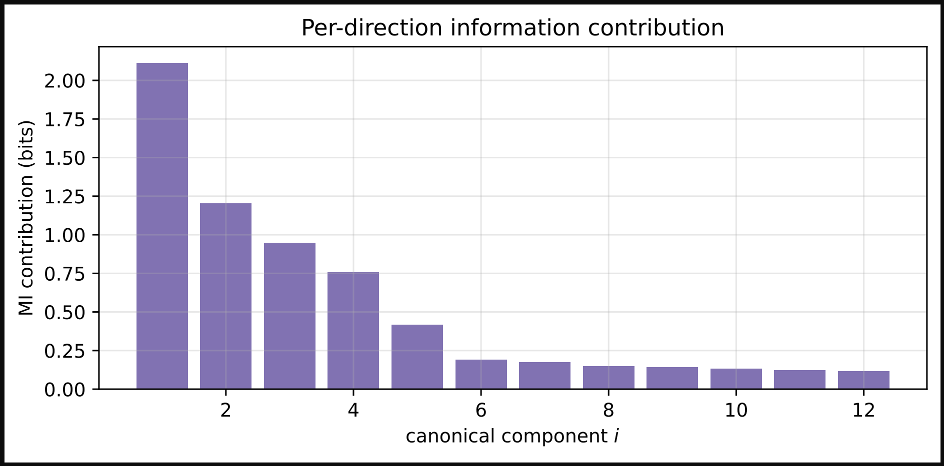


Figure 1: Per-direction contribution to $I(G;X)$ on a synthetic linear–Gaussian instance ($n = 800$, $p = 80$, $d = 40$). Each bar is $\frac{1}{2}\log(1+d_i^2)$ in bits; the dashed line is the cumulative total.

8 Canonical coordinates and the recoverability spectrum

The eigenvalues λ_i are most interpretable after whitening. Define the whitened genomic variable $\tilde{G} = \Sigma_G^{-1/2}G \sim \mathcal{N}(0, I_p)$ and the *whitened transfer operator*

$$M := \sigma^{-1}A\,\Sigma_G^{1/2} \in \mathbb{R}^{d\times p}, \qquad M = \sum_i d_i\,u_i v_i^\top \quad \text{(SVD)}, \tag{12}$$

with left/right singular vectors $u_i \in \mathbb{R}^d$, $v_i \in \mathbb{R}^p$ and singular values $d_1 \geq d_2 \geq \cdots \geq 0$.

Proposition 4 (Canonical decomposition of recoverability; recombination of Hotelling CCA 5 and probabilistic CCA 6)**.** *With M as in* (12)*, define the genomic canonical direction $a_i := \Sigma_G^{-1/2}v_i$ and the canonical score $S_i := a_i^\top G = v_i^\top\tilde{G}$. Then:*

(i) S_i *has prior variance* 1 *and posterior variance* $\mathrm{Var}(S_i \mid X) = (1 + d_i^2)^{-1}$*;*

(ii) the canonical correlation between S_i *and imaging is* $\rho_i = d_i/\sqrt{1 + d_i^2}$*, equivalently* $d_i^2 = \rho_i^2/(1 - \rho_i^2)$*;*

(iii) the eigenvalues of Section 7 satisfy $\lambda_i/\sigma^2 = d_i^2$*, so*

$$I(G; X) = \tfrac{1}{2}\sum_i \log(1 + d_i^2) = -\tfrac{1}{2}\sum_i \log(1 - \rho_i^2). \tag{13}$$

Proof. Write the SVD as $M = UDV^\top$ with $U^\top U = I$, $V^\top V = I$, and $D = \mathrm{diag}(d_1, d_2, \dots)$, so $V = [v_1, v_2, \dots]$ collects the right singular vectors and u_i the left ones. Since $S_i = v_i^\top \tilde{G}$ with $\tilde{G} \sim \mathcal{N}(0, I_p)$, the prior variance is $\mathrm{Var}(S_i) = v_i^\top v_i = 1$.

Posterior variance (i). In whitened coordinates the mean readout is $AG = A\Sigma_G^{1/2}\tilde{G} = \sigma M\tilde{G}$, so the likelihood is $X \mid \tilde{G} \sim \mathcal{N}(\sigma M\tilde{G},\ \sigma^2 I_d)$. Applying the precision form (9) to the pair $(\tilde{G}, X)$ — whose prior covariance is I_p and whose transfer matrix is σM with noise $\sigma^2 I_d$ — the posterior precision of $\tilde{G}$ is

$$\mathrm{Var}(\tilde{G} \mid X)^{-1} = I_p + \sigma^{-2}(\sigma M)^\top(\sigma M) = I_p + M^\top M = V\,(I_p + D^2)\,V^\top,$$

using $M^\top M = VD^2V^\top$. Inverting and applying $V^\top V = I_p$,

$$\mathrm{Var}(\tilde{G} \mid X) = V\,(I_p + D^2)^{-1}V^\top,$$

and therefore

$$\mathrm{Var}(S_i \mid X) = v_i^\top\,\mathrm{Var}(\tilde{G} \mid X)\,v_i = \big[(I_p + D^2)^{-1}\big]_{ii} = \frac{1}{1 + d_i^2},$$

since the v_i are orthonormal. This proves (i).

Canonical correlation (ii). The cross-covariance of the whitened genomics with imaging is

$$\mathrm{Cov}(\tilde{G}, X) = \mathbb{E}\big[\tilde{G}\,(\sigma M\tilde{G} + \varepsilon)^\top\big] = \sigma\,\mathbb{E}[\tilde{G}\tilde{G}^\top]\,M^\top + \mathbb{E}[\tilde{G}\varepsilon^\top] = \sigma M^\top,$$

using $\mathbb{E}[\tilde{G}\tilde{G}^\top] = I_p$ and $\tilde{G} \perp \varepsilon$. Hence $\mathrm{Cov}(S_i, X) = v_i^\top\,\mathrm{Cov}(\tilde{G}, X) = \sigma v_i^\top M^\top = \sigma d_i u_i^\top$, the last step because $v_i^\top M^\top = (Mv_i)^\top = (d_i u_i)^\top$ from the SVD. Pair S_i with the matched imaging variate $T_i := u_i^\top X$. From $\Sigma_X = \sigma^2(MM^\top + I_d)$ (Proposition 1 in whitened form) and $MM^\top u_i = d_i^2 u_i$,

$$\mathrm{Var}(T_i) = u_i^\top \Sigma_X u_i = \sigma^2\big(u_i^\top MM^\top u_i + 1\big) = \sigma^2(1 + d_i^2),$$

while $\mathrm{Cov}(S_i, T_i) = \mathrm{Cov}(S_i, X)\,u_i = \sigma d_i u_i^\top u_i = \sigma d_i$. The correlation is therefore

$$\rho_i = \mathrm{corr}(S_i, T_i) = \frac{\mathrm{Cov}(S_i, T_i)}{\sqrt{\mathrm{Var}(S_i)\ \mathrm{Var}(T_i)}} = \frac{\sigma d_i}{\sqrt{1}\,\sqrt{\sigma^2(1 + d_i^2)}} = \frac{d_i}{\sqrt{1 + d_i^2}},$$

and rearranging gives $d_i^2 = \rho_i^2/(1 - \rho_i^2)$, proving (ii).

Connection to the eigenvalues (iii). From $M = \sigma^{-1}A\Sigma_G^{1/2}$ we get $A\Sigma_G A^\top = \sigma^2 MM^\top$, whose nonzero eigenvalues are $\sigma^2 d_i^2$; comparing with the λ_i of Proposition 3 gives $\lambda_i = \sigma^2 d_i^2$, i.e. $\lambda_i/\sigma^2 = d_i^2$. Substituting into (11) and using $1 + d_i^2 = 1/(1 - \rho_i^2)$ from (ii),

$$I(G; X) = \tfrac{1}{2}\sum_i \log(1 + d_i^2) = -\tfrac{1}{2}\sum_i \log(1 - \rho_i^2),$$

which is (iii). □

Part (i) gives the headline identity. Define the recoverability of a genomic score as one minus the fraction of its variance that survives observing imaging (formalized in Section 9); then

$$\boxed{R(a_i) = 1 - \mathrm{Var}(S_i \mid X) = 1 - \frac{1}{1 + d_i^2} = \frac{d_i^2}{1 + d_i^2} = \rho_i^2.} \tag{14}$$

The recoverability of the i-th canonical genomic direction is exactly the squared canonical correlation. This is what reduces the whole analysis to CCA between $\mathbf{G}$ and $\mathbf{X}$ (Section 12), and it means A and σ^2 never have to be estimated individually — only the three second moments $\Sigma_G, \Sigma_X, \Sigma_{GX}$ enter.

9 Recoverability of a named genomic direction

In practice one often cares about a specific, biologically named genomic score $Y = v^\top G$ for a fixed $v \in \mathbb{R}^p$ — a pathway signature, a polygenic score, a single gene's expression ($v = e_j$).

Definition 1 (Recoverability score)**.** *For $v \in \mathbb{R}^p$ with $v^\top \Sigma_G v > 0$,*

$$R(v) := 1 - \frac{v^\top \Sigma_{G|X}\, v}{v^\top \Sigma_G\, v} = \frac{v^\top (\Sigma_G - \Sigma_{G|X})\, v}{v^\top \Sigma_G\, v} \;\in\; [0,1]. \tag{15}$$

$R(v) = 0$ means imaging carries no information about $Y = v^\top G$; $R(v) = 1$ means imaging determines it exactly. The score is scale-invariant in v and, by the Gaussian conditional-variance identity, equals the population R^2 of the Bayes-optimal predictor of Y from X: $R(v) = 1 - \mathbb{E}[\mathrm{Var}(Y \mid X)]/\mathrm{Var}(Y)$. The matrix at the core of (15) has the convenient moment-only form

$$\Sigma_G - \Sigma_{G|X} = \Sigma_G A^\top \Sigma_X^{-1} A \Sigma_G = \Sigma_{GX}\, \Sigma_X^{-1}\, \Sigma_{XG}. \tag{16}$$

10 The recoverability eigenproblem

The natural project-level question — *which* genomic directions does imaging see best — is the variational problem

$$\max_{v \neq 0} R(v) = \max_{v \neq 0} \frac{v^\top \big(\Sigma_{GX}\Sigma_X^{-1}\Sigma_{XG}\big) v}{v^\top \Sigma_G\, v}. \tag{17}$$

Proposition 5 (Recoverability eigenproblem = CCA; the generalized eigenproblem is Hotelling's 5)**.** *The stationary points of* (17) *are the solutions of the generalized eigenproblem*

$$\Sigma_{GX}\, \Sigma_X^{-1}\, \Sigma_{XG}\, v = \mu\, \Sigma_G\, v. \tag{18}$$

Its eigenvalues are $\mu_i = \rho_i^2 = d_i^2/(1+d_i^2)$ and its eigenvectors are the canonical genomic directions a_i of Proposition 4, Σ_G-orthonormal ($a_i^\top \Sigma_G a_j = \delta_{ij}$). Thus the ordered solutions of (18) *are precisely the canonical correlation directions between G and X, and $\max_v R(v) = \rho_1^2$.*

Proof. Write $C := \Sigma_{GX}\Sigma_X^{-1}\Sigma_{XG}$ for the (symmetric positive-semidefinite) numerator matrix, so the objective is the generalized Rayleigh quotient $R(v) = v^\top C v / v^\top \Sigma_G v$. Setting its gradient to zero,

$$\nabla_v R(v) = \frac{2Cv\,(v^\top \Sigma_G v) - 2\Sigma_G v\,(v^\top C v)}{(v^\top \Sigma_G v)^2} = 0 \quad\Longleftrightarrow\quad Cv = \frac{v^\top C v}{v^\top \Sigma_G v}\,\Sigma_G v = \mu\,\Sigma_G v,$$

with $\mu = R(v)$ the value of the quotient at the stationary point. This is the generalized eigenproblem (18).

To solve it, whiten via $v = \Sigma_G^{-1/2} w$. Left-multiplying (18) by $\Sigma_G^{-1/2}$ and inserting $\Sigma_G^{-1/2}\Sigma_G^{1/2} = I$ turns it into the ordinary symmetric eigenproblem

$$\big(\Sigma_G^{-1/2} C\, \Sigma_G^{-1/2}\big)\, w = \mu\, w.$$

Now expand C. Using $\Sigma_{GX} = \Sigma_G A^\top$ and $\Sigma_{XG} = A\Sigma_G$,

$$\Sigma_G^{-1/2} C\, \Sigma_G^{-1/2} = \Sigma_G^{-1/2}\,(\Sigma_G A^\top)\,\Sigma_X^{-1}\,(A\Sigma_G)\,\Sigma_G^{-1/2} = \big(\Sigma_G^{1/2} A^\top\big)\,\Sigma_X^{-1}\,\big(A\Sigma_G^{1/2}\big).$$

Substituting $A\Sigma_G^{1/2} = \sigma M$ and $\Sigma_X = \sigma^2(MM^\top + I_d)$,

$$\Sigma_G^{1/2} A^\top\, \Sigma_X^{-1}\, A\Sigma_G^{1/2} = (\sigma M^\top)\cdot \sigma^{-2}(MM^\top + I_d)^{-1}\cdot(\sigma M) = M^\top (MM^\top + I_d)^{-1} M.$$

Insert the SVD $M = UDV^\top$. Then $MM^\top + I_d = UD^2U^\top + I_d$, and on the range of U its inverse acts as $U(D^2+I)^{-1}U^\top$, so

$$M^\top (MM^\top + I_d)^{-1} M = VDU^\top \cdot U(D^2+I)^{-1}U^\top \cdot UDV^\top = V\,D^2(D^2+I)^{-1}\,V^\top,$$

using $U^\top U = I$. This is an eigendecomposition: the eigenvalues are $d_i^2/(1+d_i^2)$ with eigenvectors $w = v_i$. By Proposition 4(ii) these eigenvalues equal ρ_i^2, so $\mu_i = \rho_i^2$ and the leading value is $\max_v R(v) = \mu_1 = \rho_1^2$.

Undoing the whitening, the genomic eigenvector is $v = \Sigma_G^{-1/2} w = \Sigma_G^{-1/2} v_i = a_i$, the canonical direction of Proposition 4. Finally, the Σ_G-orthonormality follows because the v_i are orthonormal:

$$a_i^\top \Sigma_G\, a_j = (\Sigma_G^{-1/2} v_i)^\top \Sigma_G\, (\Sigma_G^{-1/2} v_j) = v_i^\top v_j = \delta_{ij}.$$

□

So the deliverable of the analysis is a *recoverability spectrum* $\rho_1^2 \geq \rho_2^2 \geq \cdots$ together with the genomic directions a_i that achieve it. Reading the loadings of $a_1, a_2, \ldots$ back onto genes or pathways answers "which biology is image-identifiable": the leading directions typically align with coarse, spatially expressed programs — proliferation, hypoxia, angiogenesis, necrosis, immune infiltration, tumor burden, stromal remodeling — rather than with individual transcripts, consistent with clear-cell RCC radiogenomic findings [24, 25, 35].

Illustration on synthetic data. Figure 2 shows the recoverability spectrum on a synthetic radiogenomic instance generated from the linear–Gaussian model of Section 4 ($n = 800$, $p = 80$, $d = 40$, latent rank $k = 6$ with prescribed $\{\rho_i^\star\}$, the last of which is near zero). The estimated $\hat{\rho}_i^2$ closely match the planted values for the identifiable directions and decay toward the high-dimensional bulk thereafter, validating Proposition 5. Figure 3 reads the same estimator against the analytic ground truth derived from Propositions 2–4: the recovered posterior mean $\hat{B}X$, posterior-variance retention, and direction alignment all track the closed-form predictions.

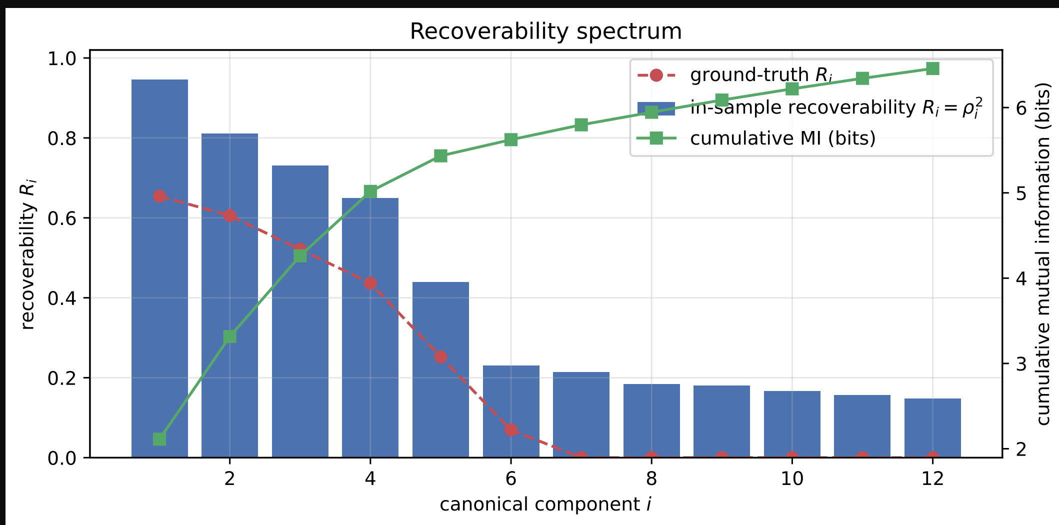


Figure 2: Recoverability spectrum on a synthetic linear–Gaussian instance ($n = 800$, $p = 80$, $d = 40$). The leading $\hat{\rho}_i^2$ recovers the planted canonical structure; later components fall to the high-dimensional bulk.

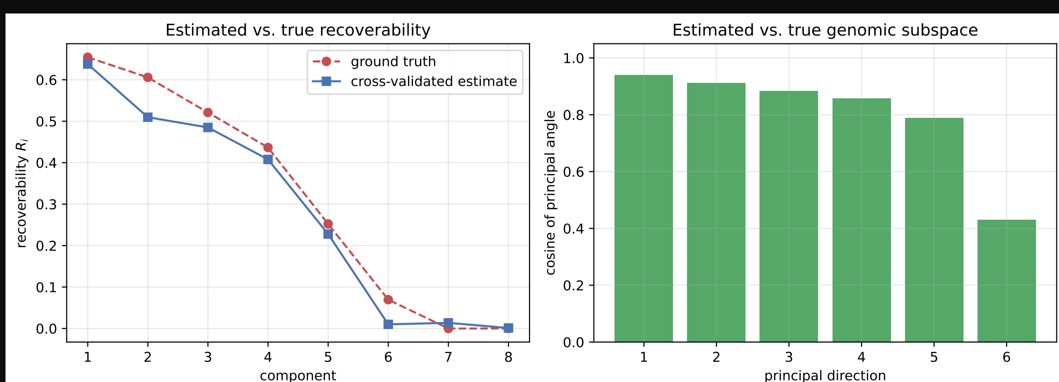


Figure 3: Empirical-vs-analytic check on the same synthetic instance. Estimated canonical correlations, recoverability per direction, and imaging variance attributable to genomics all match the closed-form quantities from Propositions 2–4.

11 Rank limits and identifiability

Proposition 6 (Rank limit)**.** *The recoverable signal matrix has rank*

$$\operatorname{rank}(\Sigma_G - \Sigma_{G|X}) = \operatorname{rank}(A\Sigma_G A^\top) \leq \min\big(d,\ p,\ \operatorname{rank} A,\ \operatorname{rank}\Sigma_G\big). \tag{19}$$

Consequently at most $\min(d, p, \operatorname{rank} A, \operatorname{rank}\Sigma_G)$ *canonical correlations are nonzero, and the image-identifiable genomic subspace* $\operatorname{span}\{a_i : \rho_i > 0\}$ *has dimension bounded by the same quantity.*

This is the conceptual core. Even with $p = 20{,}000$ measured genes, imaging can expose only a projection of genomics whose dimension is capped by the rank of the transfer map A — if a CT phenotype reflects, say, 20 macroscopic biological axes, then no model recovers 20,000 independent molecular variables from it. The recoverable dimension is a property of the *biology–imaging channel*, not of the estimator. A data-processing remark: imaging features are themselves a deterministic, lossy function $x = \phi(\text{image})$; any such post-processing can only shrink $\operatorname{rank}(\Sigma_{GX})$ and lower each ρ_i, never raise them, by the data processing inequality [8, 18, 36–38].

12 Estimation from finite samples

In applications p and d may rival or exceed n, the noise is not isotropic, and A, Σ_G, σ^2 are unknown. Because Propositions 4 and 5 require only $\Sigma_G, \Sigma_X, \Sigma_{GX}$, we estimate those directly. Figure 4 summarizes the full pipeline, from the two feature matrices through regularized covariances, the whitened cross-modal CCA matrix, its SVD, and the resulting recoverability spectrum and image-identifiable genomic directions.

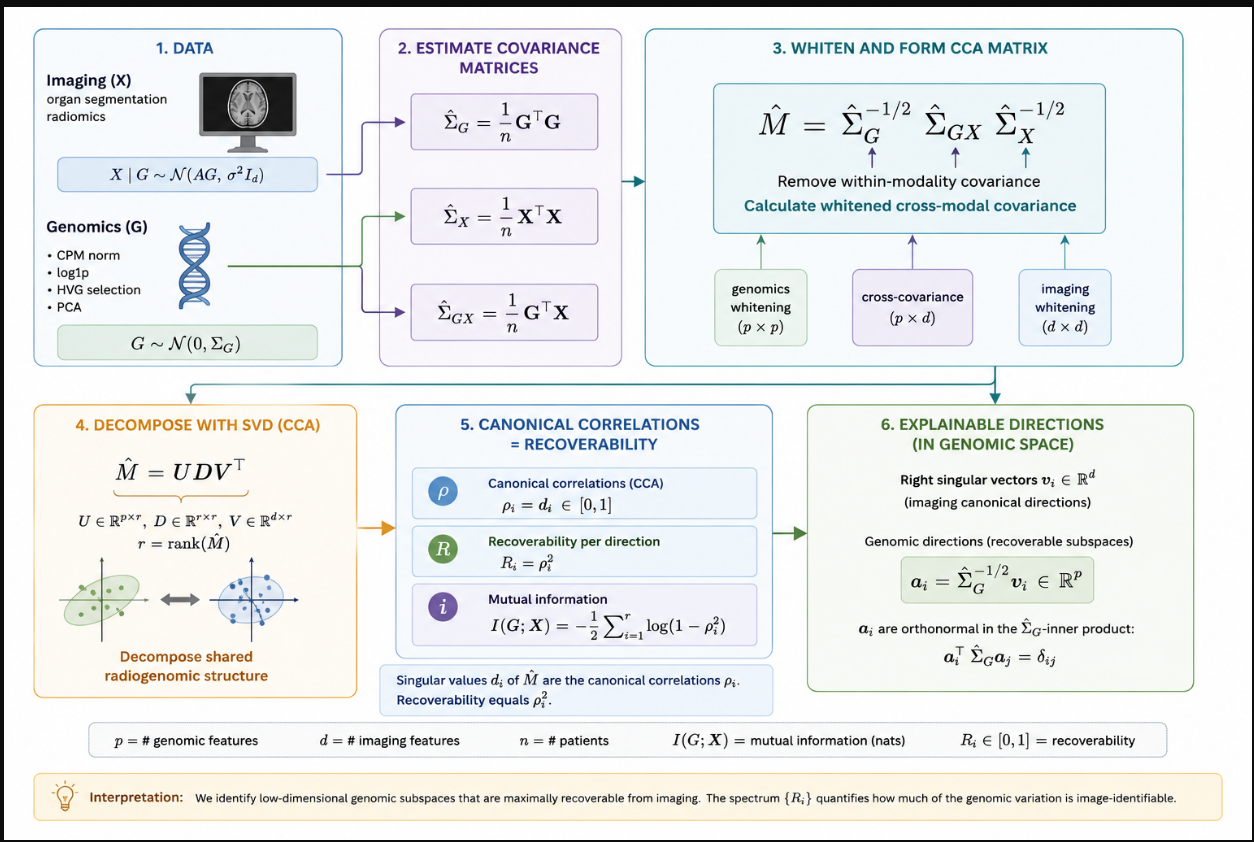


Figure 4: Overview of the recoverability estimator. From the imaging (X) and genomic (G) feature matrices we form regularized covariances $\hat\Sigma_G, \hat\Sigma_X, \hat\Sigma_{GX}$, whiten and assemble the cross-modal matrix $\hat M = \hat\Sigma_G^{-1/2}\hat\Sigma_{GX}\hat\Sigma_X^{-1/2}$, and take its SVD. The singular values are the canonical correlations $\hat\rho_i$; the recoverabilities are $\hat R_i = \hat\rho_i^2$ and the mutual information is $I(G;X) = -\frac{1}{2}\sum_i \log(1-\hat\rho_i^2)$; the left singular vectors map back to image-identifiable genomic directions $\hat a_i$. The spectrum $\{\hat R_i\}$ quantifies how much of the genomic variation is identifiable from imaging and along how many directions.

12.1 Regularized covariances

Replace sample covariances by shrinkage estimators (e.g. Ledoit–Wolf [12]), $\hat\Sigma_G = (1-\alpha_G)\,S_G + \alpha_G\,\nu_G I_p$ and likewise $\hat\Sigma_X$, with $S_G = n^{-1}\mathbf{G}^\top\mathbf{G}$ the centered sample covariance; the cross-covariance is $\hat\Sigma_{GX} = n^{-1}\mathbf{G}^\top\mathbf{X}$. Shrinkage keeps $\hat\Sigma_G, \hat\Sigma_X$ invertible and well-conditioned, which is required for the whitening in (12). When p is very large it is practical to first project genomics onto its top principal components (or onto curated pathway scores), reducing

p to a manageable $p' \ll n$ before estimation; the rank limit of Proposition 6 means little is lost provided p' exceeds the recoverable dimension.

Gaussian-copula marginals. The model's Gaussian prior constrains the *marginals* of each feature, not only their dependence. Radiomic descriptors in particular are badly non-Gaussian — heavy-tailed shape and texture statistics with excess kurtosis in the tens (Section 15) — so the raw second moments $\hat{\Sigma}_G, \hat{\Sigma}_X$ are dominated by a few outlying patients and the whitening in (12) is unstable. A simple and assumption-faithful remedy is the *Gaussian-copula marginal transform*: replace each feature by its rank mapped through the inverse normal CDF, $\tilde{M}_{ij} = \Phi^{-1}\big(\mathrm{rank}_i(M_{\cdot j})/(n+1)\big)$, so every marginal is exactly standard normal and only the dependence structure (hence the canonical correlations) survives. This is the continuous-feature analogue of the latent-Gaussian construction of Section 14 and is the default applied to both modalities in the analyses below.

12.2 Reduced-rank and ridge regression

The transfer map, if needed explicitly, is the multivariate regression of X on G. Ridge gives $\hat{A}^\top = (\mathbf{G}^\top \mathbf{G} + \gamma I_p)^{-1} \mathbf{G}^\top \mathbf{X}$, and $\hat{\sigma}^2$ follows from the residual covariance. Imposing $\mathrm{rank}(A) \le k$ (reduced-rank regression) bakes Proposition 6 into the fit and is equivalent, up to regularization, to truncating the SVD of M at k components.

12.3 Regularized CCA

The estimator used in the companion notebook solves (18) with the regularized covariances of Section 12.1: form $\hat{M} = \hat{\Sigma}_G^{-1/2} \hat{\Sigma}_{GX} \hat{\Sigma}_X^{-1/2}$, take its SVD $\hat{M} = \hat{U}\hat{D}\hat{V}^\top$, and read off $\hat{\rho}_i = \hat{d}_i$ (clipped to $[0, 1)$), $\hat{R}_i = \hat{\rho}_i^2$, genomic directions $\hat{a}_i = \hat{\Sigma}_G^{-1/2} \hat{u}_i$, imaging directions $\hat{b}_i = \hat{\Sigma}_X^{-1/2} \hat{v}_i$. The Bayes-optimal predictor and posterior covariance follow from (7)–(8) with hats throughout. This is exactly regularized CCA — the recoverability program and CCA are the same computation.

12.4 Probabilistic CCA: the latent-variable view

The rank limit motivates an explicit low-dimensional generative form. With a shared latent $Z \in \mathbb{R}^k$,

$$Z \sim \mathcal{N}(0, I_k), \qquad G = W_G Z + \mu_G + \epsilon_G, \qquad X = W_X Z + \mu_X + \epsilon_X, \tag{20}$$

$\epsilon_G \sim \mathcal{N}(0, \Psi_G)$, $\epsilon_X \sim \mathcal{N}(0, \Psi_X)$ independent (probabilistic CCA; 6). Here k *is* the dimension of the image-identifiable genomic subspace, Z is the shared biological state, and ϵ_G, ϵ_X are modality-private variation. The implied moments are $\Sigma_G = W_G W_G^\top + \Psi_G$, $\Sigma_X = W_X W_X^\top + \Psi_X$, $\Sigma_{GX} = W_G W_X^\top$; the maximum-likelihood loadings are the canonical directions scaled by the canonical correlations, recovering Section 12.3. The model is fit by EM, which also yields posterior latent estimates $\mathbb{E}[Z \mid x]$ and naturally accommodates anisotropic (per-feature) noise, unlike the isotropic $\sigma^2 I_d$ of the base model.

12.5 Sample-size regimes and dimensional collapse

Equations (7)–(18) are population identities; the estimators of Sections 12.1–12.4 inherit a finite-sample bias whose magnitude is governed entirely by the aspect ratios p/n and d/n. The bias is not benign, and below a critical sample size the recoverability spectrum breaks down in a specific, predictable way.

Three regimes. Let $p^\star, d^\star$ be the working dimensions after any pre-reduction (e.g. PCA), and write $\kappa_G = p^\star/n$, $\kappa_X = d^\star/n$.

1. *Classical* ($\kappa_G + \kappa_X \ll 1$). Sample CCA is consistent; $\hat{\rho}_i \to \rho_i$ at the usual $n^{-1/2}$ rate and the in-sample spectrum is a faithful estimate of the population one.
2. *High-dimensional* ($\kappa_G + \kappa_X \lesssim 1$, both bounded away from 1). Under the null $\Sigma_{GX} = 0$, the squared sample canonical correlations follow a Wachter-type law on $[(\sqrt{\kappa_G(1-\kappa_X)} - \sqrt{\kappa_X(1-\kappa_G)})^2, (\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2]$ [9, 10]; the leading sample value already sits well above zero by geometry. Estimates of ρ_i are systematically inflated and ordinary CCA must be replaced by its regularized form, with shrinkage parameters tuned by cross-validation.
3. *Saturating* ($p^\star + d^\star \ge n$). The rows of $\mathbf{G}$ and $\mathbf{X}$ live in subspaces whose intersection has dimension at least $p^\star + d^\star - n + 1$; consequently $\hat{\rho}_1 = \cdots = \hat{\rho}_q = 1$ for $q = \min(p^\star, d^\star) - (n - \max(p^\star, d^\star))_+$

regardless of any underlying association. The empirical mutual information diverges, the in-sample $\mathrm{tr}(\hat{\Sigma}_{GX}\hat{\Sigma}_X^{-1}\hat{\Sigma}_{XG})/\,\mathrm{tr}(\hat{\Sigma}_X)$ approaches the rank fraction of the genomic span, and the permutation null collapses onto the observed value (its p-values concentrate near 1). Nothing in the spectrum is estimable in this regime without shrinkage *plus* dimension reduction; shrinkage alone keeps the matrices invertible but does not restore identifiability.

Quantitative warning signs. Three diagnostics, jointly, separate real signal from dimensional artefact:

- the gap between in-sample $\hat{R}_i$ and the K-fold value $\hat{R}_i^{\mathrm{cv}}$; under regime 3 this gap approaches $\hat{R}_i - 0$;
- the permutation null for $\hat{\rho}_1$: if the observed value lies inside the bulk, the result is consistent with chance;
- the cross-validated R^2 of the leading canonical score $a_1^\top g$ predicted from x on a held-out split; in regime 3 it is typically large and *negative* (worse than predicting the mean), a behaviour seen empirically on TCGA-KIRC at $n = 131$ with $p^\star = 100, d^\star = 107$ in the companion notebook.

Recommendations. Choose $p^\star$ and $d^\star$ *adaptively from* n, not from the native modality dimensions:

- Target $p^\star + d^\star \leq n/\tau$ with $\tau \geq 5$ as a working rule (we use $p^\star, d^\star \leq \lfloor n/\tau \rfloor$ per side in the notebook; $\tau \in [5, 10]$ trades bias for resolution). Reduce both modalities by PCA when needed — reducing only the larger side leaves the estimator in regime 3.
- Treat $\hat{R}_i^{\mathrm{cv}}$ (Section 12.6) as the primary estimand and report the in-sample $\hat{R}_i$ only with the gap made explicit; the in-sample fraction of imaging variance attributable to genomics should be reported alongside a cross-validated counterpart, e.g. $\sum_i \hat{R}_i^{\mathrm{cv}}/\min(p^\star, d^\star)$.
- Power planning. The leading population correlation ρ_1 is detectable above the Wachter bulk only when its squared value exceeds $(\sqrt{\kappa_G(1-\kappa_X)} + \sqrt{\kappa_X(1-\kappa_G)})^2$; solving for n gives a minimum sample size given a hypothesised effect. For $\rho_1^2 = 0.2$ and a balanced $p^\star = d^\star$, this requires roughly $n \gtrsim 25\,p^\star$, an order of magnitude beyond what cohort sizes in radiogenomics usually offer — reinforcing the need for aggressive pre-reduction.
- Probabilistic CCA (Section 12.4) with a small, fixed latent dimension k is the most parsimonious option when n is the binding constraint; rank-k regularization sidesteps the high-dimensional bulk by construction.

The practical message is conservative: in radiogenomic cohorts, sample size is almost always the binding constraint and a defensible analysis is the one whose effective $p^\star + d^\star$ is a small fraction of n, with both sides reduced symmetrically and every reported number a cross-validated or permutation-calibrated one.

Figure 5 illustrates these regimes on the synthetic instance, where the planted truth is known. As n grows with $p^\star, d^\star$ scaled to keep $\kappa_G + \kappa_X$ bounded, the in-sample $\hat{\rho}_1$ sits at or near 1 across the small-n saturating regime — indistinguishable from the permutation null — while the cross-validated $\hat{R}_1^{\mathrm{cv}}$ only separates from zero once n clears the Wachter edge. The empirical mutual information shows the same threshold behaviour.

The sweep above grows $p^\star, d^\star$ with n, which entangles “more data” with “bigger model”. The complementary question — how much data is needed at a *fixed* model — is answered by Figure 6, which holds $p^\star = d^\star = 20$ and varies only n on the same synthetic instance, whose planted leading recoverability $\rho_1^2 = 0.65$ is known in closed form (Proposition 5). Two thresholds separate. The cross-validated *point* estimate $\hat{R}_1^{\mathrm{cv}}$ is already close to the truth at small n (within 90% by $n \approx 80$, i.e. $n \approx 2(p^\star + d^\star)$), because the planted signal is strong. But the estimate cannot be *certified* against the permutation null until later: the null 95% quantile is inflated to ≈ 1 in the saturating regime ($n \lesssim p^\star + d^\star$) and only decays below the estimate at $n \approx 176 \approx 4.4(p^\star + d^\star)$. The practical reading is that “how much data” depends on the claim: recovering the right number takes far less data than proving it is not an artefact of the high-dimensional geometry, and the cohort downsample sweeps (Sections 17.5) report the latter, conservative threshold.

12.6 Honest recoverability: cross-validation and permutation

In-sample canonical correlations are biased upward, severely so when p, d are not small relative to n. Two safeguards, both implemented in the notebook:

- *Cross-validation.* Estimate $\hat{a}_i, \hat{b}_i$ on a training fold; on the held-out fold form the scores $\hat{a}_i^\top g$ and $\hat{b}_i^\top x$ and report their out-of-sample squared correlation as the honest $\hat{R}_i$. The gap to the in-sample value quantifies overfitting.

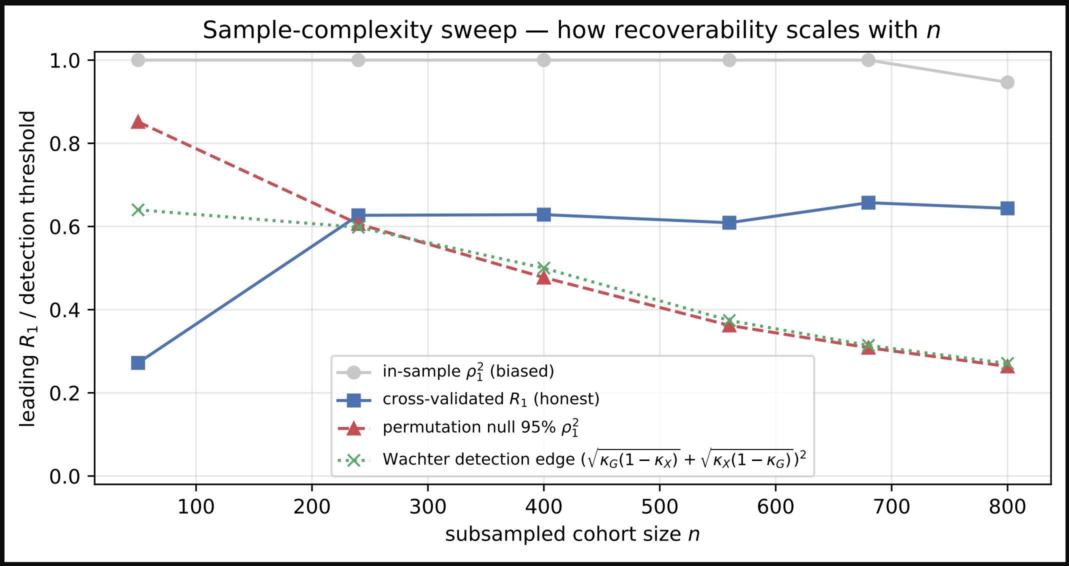


Figure 5: Sample-complexity sweep on the synthetic instance with known truth. In-sample $\hat{\rho}_1$ tracks the null until n exceeds the Wachter edge; cross-validated $\hat{R}_1^{\mathrm{cv}}$ and empirical $\hat{I}(G;X)$ only become informative past that threshold. Compare to Figure 12c for the same diagnostic on TCGA-KIRC.

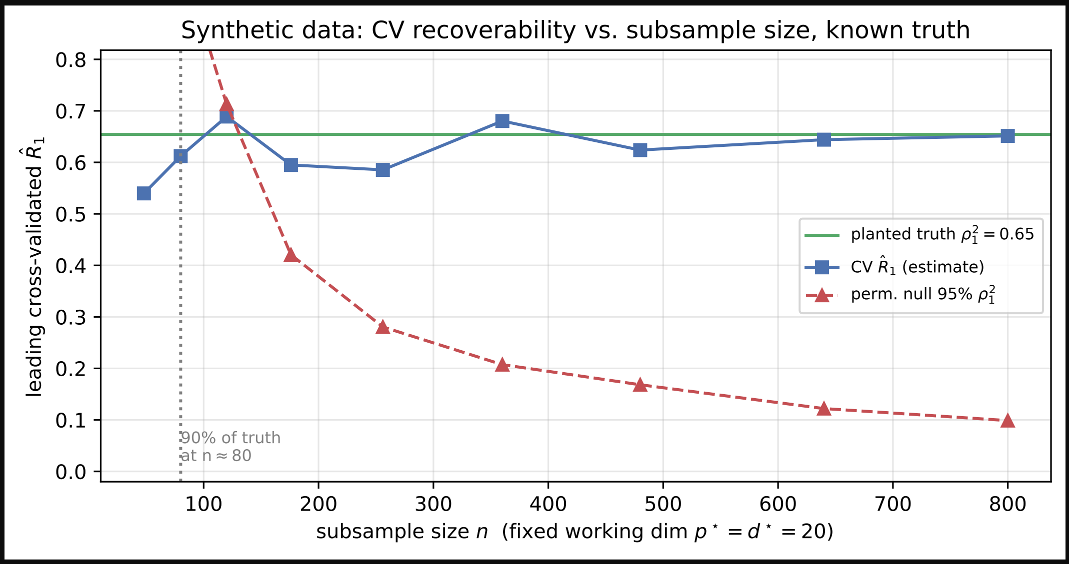


Figure 6: Fixed-model downsampling sweep on the synthetic instance ($p^\star = d^\star = 20$, planted $\rho_1^2 = 0.65$). The cross-validated estimate (blue) tracks the known truth (green) even at small n, but the permutation null 95% quantile (red, dashed) is saturated near 1 in the small-n regime and only falls below the estimate at $n \approx 176$. Recovering the value needs less data than certifying it. Companion to the growing-model sweep of Fig. 5 and the cohort sweeps (Fig. 16e).

- *Permutation test.* Repeatedly permute the patient labels of one modality, refit, and collect the leading canonical correlation to build a null distribution; the permutation p-value tests whether any genomic direction is image-identifiable beyond chance.

Scale-matched effective rank. A subtlety arises when these two safeguards are combined to count *how many* directions are identifiable. The in-sample permutation null of the leading $\hat{\rho}_1$ is itself inflated toward 1 by the high-dimensional geometry of Section 12.5, whereas $\hat{R}_i^{\mathrm{cv}}$ is honestly deflated; thresholding the deflated statistic against the inflated null forces the estimated rank to zero regardless of any true signal. The two must be compared on the same scale. We therefore build the null of the *cross-validated* statistic directly: permute patient labels, recompute the full K-fold $\hat{R}_i^{\text{cv}}$, and define the *effective image-identifiable rank* as the number of directions whose observed $\hat{R}_i^{\text{cv}}$ exceeds its own permutation 95% quantile at permutation $p < 0.05$. This is the estimand we report below.

Figure 7 shows both diagnostics on the synthetic instance. The in-sample $\hat{R}_i$ overshoots the planted truth by the amount predicted by the high-dimensional bias, while the 5-fold $\hat{R}_i^{\text{cv}}$ tracks the truth for identifiable directions and collapses toward the null 95% quantile thereafter. Figure 8 shows the permutation null on $\hat{\rho}_1$: when the true signal is nonzero, the observed value sits cleanly outside the null bulk and the per-direction p-values fall below 0.01 for the top three components, matching the planted rank.

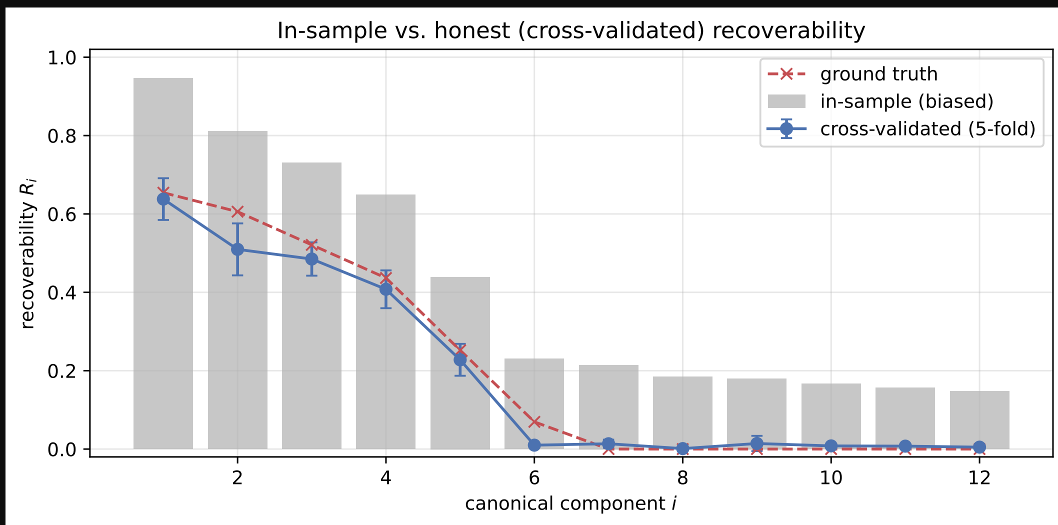


Figure 7: Cross-validated recoverability on a synthetic instance with planted rank $k = 6$ (the sixth correlation near zero). In-sample $\hat{R}_i$ (grey) is biased upward; 5-fold $\hat{R}_i^{\text{cv}}$ (blue) tracks the planted truth for the identifiable directions and collapses to the null floor afterwards.

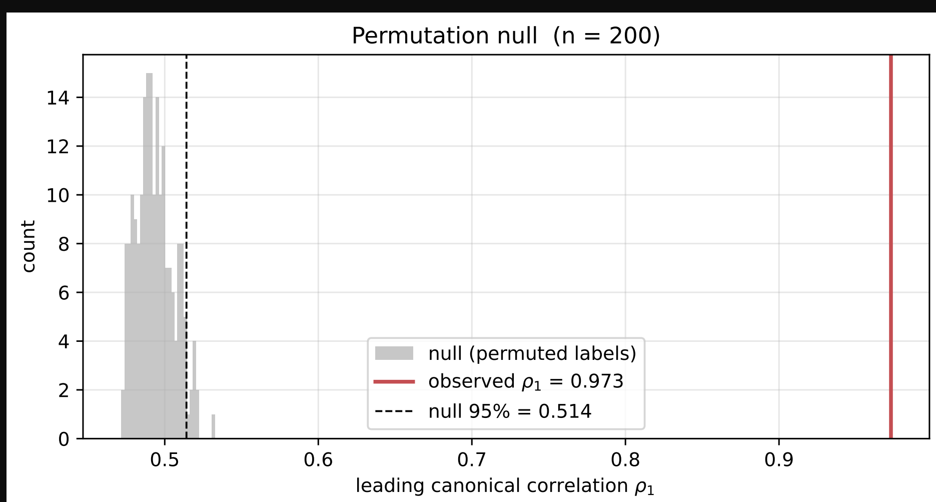


Figure 8: Permutation null on the leading canonical correlation $\hat{\rho}_1$ for the synthetic instance. Red marks the observed value; $p < 0.01$ for the top three directions, recovering the planted rank.

The same optimism appears in the aggregate fraction of imaging variance attributable to genomics. Figure 9 shows the in-sample share $\sum_i \hat{R}_i / \min(p^\star, d^\star)$ alongside its cross-validated counterpart on the synthetic instance: the in-sample number sits well above the planted truth while the CV value tracks it. The same diagnostic on TCGA-KIRC (Section 15) reports 0.193 in-sample against 0.012 cross-validated — a sixteen-fold gap that is consistent with the synthetic mechanism rather than with any genuinely recoverable signal.

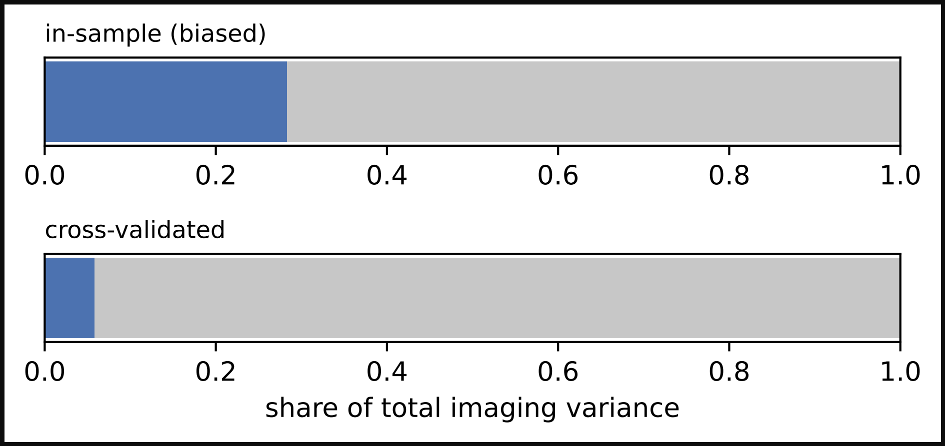


Figure 9: Share of imaging variance attributable to genomics on the synthetic instance: in-sample (grey) versus 5-fold cross-validated (blue), with the planted truth marked. The in-sample share is biased upward by the high-dimensional inflation described in Section 12.5.

12.7 Checking the model assumptions

The closed-form recoverability and mutual information are exact *under* the linear–Gaussian law; before trusting the numbers we verify, upfront, how far each cohort departs from it. Three checks bracket the assumptions the estimator actually uses.

(i) Marginal and joint Gaussianity. The Gaussian-copula transform (Section 12.1) makes every *marginal* exactly normal by construction; the question is whether the *joint* $[G_{\text{pca}}, X_{\text{pca}}]$ is. We test it with a Mahalanobis-distance QQ plot against the χ^2_p law (squared Mahalanobis distances are χ^2_p under joint MVN) and Mardia's standardized skew/kurtosis ratios (Fig. 10). Per-modality marginal normality does not imply joint normality, so this is the assumption the spectrum genuinely relies on.

(ii) Linearity of the dependence. Canonical correlation sees only second-order (linear) coupling. We complement it with the distance correlation [39], which is zero *iff* G_{pca} and X_{pca} are independent and therefore detects non-linear and higher-order dependence the linear–Gaussian estimator is blind to. We report a permutation p-value (the statistic has a positive finite-sample bias, so its null, not its raw value, is the reference). A significant distance correlation where the cross-validated recoverability is null would indicate a channel the linear bound understates.

(iii) Per-feature marginals. As documentation of the copula step, we summarize per-feature skewness and excess kurtosis on the raw features and confirm both vanish after the transform.

Findings. Table 1 collects the three checks across cohorts. Two patterns are robust. First, the copula transform is doing real work: raw genomic marginals are severely non-Gaussian (78% of KIRC probes fail $|\text{skew}| \leq 2$, $|\text{kurt}| \leq 7$, median excess kurtosis 70), and all go to 0% afterwards. Second, and less obviously, *marginal Gaussianity does not buy joint Gaussianity*: even post-copula, the share of samples beyond the χ^2 99% shell is 7–16% rather than the nominal 1%. Part is finite-sample (the ratio $n/(\dim G + \dim X) \approx 2.5$ in all three inflates the Mahalanobis tail), but it shrinks monotonically with n ($16\% \to 12\% \to 8\%$ for KIRC $\to$ NSCLC $\to$ ADNI), implicating genuine residual *tail dependence* the copula cannot remove. The Mardia skew ratio stays ≈ 0, so the joint is symmetric but leptokurtic. The distance correlation is significant in every cohort ($p = 0.005$) but only modestly above its inflated null, and its ordering tracks the *linear* recoverability rather than exceeding it — there is no cohort where non-linear dependence lights up while the linear channel is dead.

What this does and does not cost the upper bound. It is worth separating two claims that the phrase "$I(G; X)$ is an upper bound" can mean. The *operational* bound — that no estimator, linear or non-linear, can recover G from X better than $I(G; X)$ allows — is the data processing inequality $G \to X \to \hat{g}(X)$ and holds for the *true* joint law regardless of its shape (Section 14.2). What requires Gaussianity is the second, computational claim: that the *number* we report, $-\frac{1}{2}\sum_i \log(1 - \hat{\rho}_i^2)$, equals that true $I(G; X)$. The copula secures the marginals; the residual joint non-Gaussianity above means the two are equal only approximately. Crucially, the mis-specification is one-sided in a

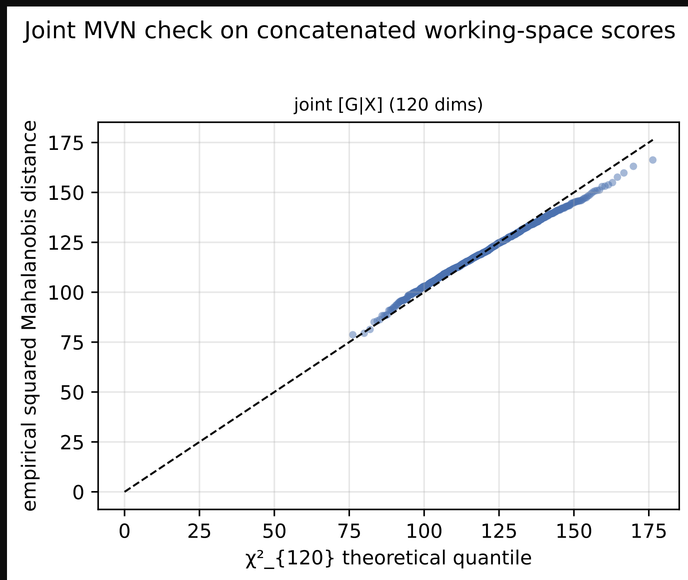
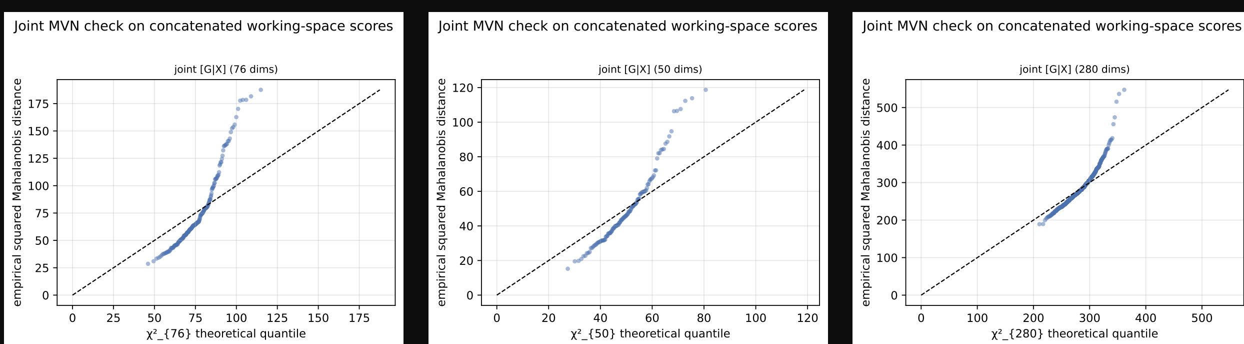


Figure 10: Joint-MVN check on the concatenated working-space scores $[G_{\mathrm{pca}}, X_{\mathrm{pca}}]$ for the synthetic instance: squared Mahalanobis distances against χ^2_p quantiles lie on the line, with kurtosis ratio ≈ 1 and a 0.4% tail beyond the 99% shell. The real cohorts (Fig. 11) depart from this in their upper tails.



(a) TCGA-KIRC (b) NSCLC (c) ADNI

Figure 11: Joint-MVN QQ plots on the three real cohorts (squared Mahalanobis distances of $[G_{\mathrm{pca}}, X_{\mathrm{pca}}]$ vs. χ^2_p quantiles; the line $y = x$ is exact joint Gaussianity). The copula makes every marginal Gaussian, so departures here are purely *joint* non-Gaussianity. All three track the line through the bulk and lift above it in the upper tail — the residual leptokurtosis (heavy joint tails) quantified in Table 1. The deviation is largest in the small tumour cohorts (a, b) and mildest in ADNI (c), the cohort with the most favourable $n/(\dim G + \dim X)$ ratio, consistent with the part of the inflation that is finite-sample. Together with the synthetic ideal (Fig. 10) these panels are the visual companion to the assumption table.

favourable direction: because canonical correlation captures only linear coupling, ignoring higher-order dependence can only make the Gaussian formula *under*-count the true $I(G;X)$, never over-count it — so the reported figure is conservative as a statement about the true channel, not optimistic. The distance-correlation check bounds how much is being missed (little), and the finite-sample optimism that runs the *other* way is removed separately by the permutation-calibrated upper confidence limit (Section 15.4). The net reading: significance and the identifiable-rank conclusions are distribution-free and stand; the MI *magnitudes* are best read as approximate, most trustworthy in ADNI (near-Gaussian joint, large n) and most caveated in the small, heavier-tailed tumour cohorts.

A signed bias, made precise. The favourable direction is not heuristic; the copula pins it down. Write $I(G;X) = h(G) + h(X) - h(G,X)$. The copula forces each marginal to be exactly Gaussian, so $h(G)$ and $h(X)$ equal their Gaussian values and the *only* term in which the true law can differ from the Gaussian surrogate is the joint entropy

	TCGA-KIRC	NSCLC	ADNI
n / joint dim	$190/76$	$129/50$	$702/280$
Joint kurt. ratio (1 = MVN)	1.17	1.14	1.02
Tail beyond $\chi^2_{.99}$ (1% = MVN)	15.8%	12.4%	8.1%
Genomic marginals non-normal (raw → post)	$78\% \to 0\%$	$57\% \to 0\%$	$24\% \to 0\%$
Distance corr. (obs / null$_{95}$ / p)	$0.43/0.37/.005$	$0.34/0.31/.005$	$0.32/0.27/.005$

Table 1: Model-assumption diagnostics. The copula removes all marginal non-normality; the joint remains mildly heavy-tailed (worst in the small tumour cohorts, near-Gaussian in ADNI); the distance correlation reveals no non-linear channel beyond the linear one.

$h(G, X)$. Because the Gaussian is the maximum-entropy law for a fixed covariance, $h(G, X) \le h_{\text{gauss}}(G, X)$, whence

$$I(G; X) - \left(-\tfrac{1}{2}\textstyle\sum_i \log(1 - \rho_i^2)\right) \;=\; h_{\text{gauss}}(G, X) - h(G, X) \;\ge\; 0, \tag{21}$$

with equality *iff* the joint is Gaussian. So, given Gaussian marginals, the population canonical-correlation mutual information is a genuine *lower* bound on the truth — exact under joint Gaussianity, an under-estimate otherwise — and the gap is precisely the non-Gaussian (non-linear, higher-order) dependence that CCA cannot see, which is what the distance correlation probes. (Without the Gaussian-marginal condition the marginal negentropies re-enter (21) and the sign is no longer guaranteed; securing this clean direction is a concrete reason to apply the copula.) The two error sources therefore push in *opposite* directions: population mis-specification can only *lower* the estimate relative to the true $I(G; X)$, while finite-sample inflation of $\hat{\rho}_i$ *raises* it. The permutation-calibrated upper confidence limit (Section 15.4) controls the second; (21) and the distance-correlation check bound the first. A reported figure that is small after both controls is therefore a trustworthy statement that the channel is genuinely narrow, not an artefact of either bias.

13 Interpretation for radiogenomics

The model reframes the radiogenomic question [1–4]. Rather than testing whether CT predicts each gene — a high-dimensional, low-power multiple-testing problem — it asks for the *recoverability spectrum* $\{\rho_i^2\}$ and the associated genomic directions $\{a_i\}$. The outputs are: (1) a small number of image-identifiable genomic axes, with gene/pathway loadings that make them biologically legible; (2) a hard bound $\Sigma_{G|X}$ on what any predictor can achieve, per direction; (3) a total information budget $I(G; X)$; and (4) an honest, cross-validated estimate of each. The governing principle is that radiogenomics should target *image-identifiable low-dimensional genomic subspaces*, whose dimension is set by the biology–imaging channel and not by the gene panel size.

14 A Bernoulli extension for mutation data via a Gaussian copula

The linear–Gaussian model of Section 3 is well matched to continuous, approximately log-normal genomic readouts: log-CPM gene expression, methylation β- or M-values, log-CNV ratios. It is plainly misspecified for somatic mutation calls, where each $G_j \in \{0, 1\}$ indicates the presence of a non-silent variant in gene j and the marginal frequencies p_j are small, gene-specific, and highly heterogeneous. This section extends the framework to such data by replacing the Gaussian prior on G with a *Gaussian-copula Bernoulli* (equivalently, multivariate probit) prior, while preserving the canonical-correlation / mutual-information machinery of Sections 7–8.

14.1 The latent-Gaussian generative model

Let $p_j = \Pr(G_j = 1)$ and $\tau_j = \Phi^{-1}(1 - p_j)$, where Φ is the standard normal CDF. Introduce a latent vector

$$Z \sim \mathcal{N}(0, \Sigma_Z), \qquad \operatorname{diag}(\Sigma_Z) = I_p,$$

and define the observed mutation indicator by thresholding:

$$G_j = \mathbb{1}\{Z_j > \tau_j\}, \qquad j = 1, \ldots, p.$$

By construction the marginals are correct, $G_j \sim \operatorname{Bern}(p_j)$, and the pairwise joint laws are governed by the tetrachoric correlation $\Sigma_{Z,jk}$, which can be any positive-definite correlation matrix — providing arbitrary dependence structure between mutated genes (e.g. mutual exclusivity, co-occurrence within pathways).

Imaging is coupled to the *latent* state, not to the indicators directly:

$$X \mid Z \sim \mathcal{N}\big(B\,Z,\,\sigma^2 I_d\big), \qquad B \in \mathbb{R}^{d\times p}. \tag{22}$$

This is the natural specification: the biological mechanism that produces an imageable phenotype (proliferation, vascularity, necrosis) responds to an underlying continuous burden, of which an observed mutation is a coarse, binarized readout. The joint structure factors as

$$Z \to G, \qquad Z \to X,$$

so G and X are conditionally independent given Z.

14.2 Mutual information and a tight DPI bound

Because G is a deterministic, memoryless function of Z, the data processing inequality yields the upper bound

$$I(G;X) \;\le\; I(Z;X), \tag{23}$$

with equality only if Z can be recovered from G (impossible whenever any $p_j \in (0,1)$, since thresholding is lossy). The right-hand side reduces exactly to the Gaussian formula of Section 7: if $\{\rho_i\}_{i=1}^k$ denote the canonical correlations of (Z,X) under (22), then

$$I(Z;X) \;=\; -\tfrac{1}{2}\sum_{i=1}^{k}\log\big(1-\rho_i^2\big), \qquad k = \operatorname{rank}\big(\Sigma_Z^{1/2}B^\top\Sigma_X^{-1}B\Sigma_Z^{1/2}\big), \tag{24}$$

with $\Sigma_X = B\Sigma_Z B^\top + \sigma^2 I_d$. The recoverability spectrum, the directions $\{a_i\}$, and the eigenproblem of Section 10 all transfer verbatim, with Σ_G replaced by Σ_Z.

The binarization gap. The slack in (23) is

$$I(Z;X) - I(G;X) \;=\; H(Z \mid G) - H(Z \mid G, X) \;=\; I(Z;X \mid G),$$

the residual information about Z not already revealed by the mutation call. For a single gene with frequency p_j, the entropy lost to binarization is $H(Z_j) - H(G_j) = \frac{1}{2}\log(2\pi e) - h_2(p_j)$ where h_2 is the binary entropy in nats; rare mutations ($p_j \ll 1/2$) are highly lossy, common ones near $p_j = 1/2$ less so. Thus (23) is tightest for cohorts dominated by intermediate-frequency mutations and loosest for rare-variant panels — a useful guide to when the bound is informative.

14.3 Per-gene recoverability and AUC ceilings

For a named mutation G_j, the Bayes-optimal predictor from X is the posterior probability

$$\pi_j(x) \;=\; \Pr(G_j = 1 \mid X = x) \;=\; \Phi\!\left(\frac{\mu_j(x) - \tau_j}{s_j}\right),$$

where $(\mu_j(x), s_j^2) = (\mathbb{E}[Z_j \mid X = x],\ \operatorname{Var}(Z_j \mid X = x))$ are the linear-Gaussian posterior moments for the latent coordinate, computed from (Σ_Z, B, σ^2) exactly as in Section 6. The *latent recoverability* $R_j^Z = \operatorname{Var}(\mathbb{E}[Z_j \mid X])/\operatorname{Var}(Z_j)$ is the direct analogue of $R(v)$ in the Gaussian case, and the binary AUC of any predictor $\hat{G}_j(x)$ is bounded by

$$\mathrm{AUC}_j \;\le\; \Phi\!\left(\sqrt{\tfrac{R_j^Z}{1-R_j^Z}}\Big/\sqrt{2}\right), \tag{25}$$

attained by the Bayes classifier $\pi_j(x) > p_j$. Equation (25) is the natural object to compare against the empirical per-gene $\mathrm{AUC}_{\mathrm{LR}}$ and $\mathrm{AUC}_{\mathrm{RF}}$ of Section 15.7: an empirical AUC near 0.5 is consistent with either a small R_j^Z *or* a degenerate p_j, and (25) disentangles the two.

14.4 Estimation

The estimator splits into three independent pieces, each well-studied:

(i) **Marginal thresholds.** Estimate $\hat{p}_j = \bar{G}_{\cdot j}$ and set $\hat{\tau}_j = \Phi^{-1}(1-\hat{p}_j)$. Apply a continuity correction or empirical-Bayes shrinkage for rare mutations.

(ii) **Latent correlation $\hat{\Sigma}_Z$.** Estimate pairwise tetrachoric correlations from $2{\times}2$ tables of (G_j, G_k) by maximum likelihood under the bivariate-normal copula, then project the resulting matrix to the nearest positive-definite correlation matrix (e.g. Higham's `nearcorr` [40]). For high-dimensional panels apply Ledoit–Wolf shrinkage as in Section 12.1.

(iii) **Cross-modal covariance $\hat{\Sigma}_{ZX}$.** The point-biserial covariance $\mathrm{Cov}(G_j, X_\ell)$ is related to the latent covariance by

$$\mathrm{Cov}(G_j, X_\ell) \;=\; \phi(\tau_j)\;\mathrm{Cov}(Z_j, X_\ell),$$

where ϕ is the standard normal density. Estimate $\mathrm{Cov}(G_j, X_\ell)$ from the sample and rescale by $\phi(\hat{\tau}_j)$ to obtain $\hat{\Sigma}_{ZX,j\ell}$.

With $(\hat{\Sigma}_Z, \hat{\Sigma}_{ZX}, \hat{\Sigma}_X)$ in hand, the regularized CCA of Section 12.3, the permutation null of Section 12.6, and the empirical MI bound (estimated, e.g., by k-nearest-neighbour or neural estimators [20, 21, 41]) carry over without further change. The dimensional-collapse warnings of Section 12.5 are if anything *more* acute, since tetrachoric estimation has higher variance than Pearson correlation, particularly for rare-rare gene pairs.

14.5 Mixed-modality genomics

In practice a single cohort carries both continuous (expression, CNV) and binary (mutation) features. The copula formulation accommodates this by partitioning the latent vector as $Z = (Z^{\mathrm{cts}}, Z^{\mathrm{bin}})$ with a single joint Σ_Z. The continuous block uses the identity link ($G^{\mathrm{cts}} = Z^{\mathrm{cts}}$ up to marginal standardization) and the binary block uses the probit link of Section 14.1. Estimation of Σ_Z proceeds blockwise — Pearson within the continuous block, tetrachoric within the binary block, polyserial across blocks — after which the rest of the pipeline is identical. This is the route by which the present manuscript's gene-expression analysis (Section 15) and the per-gene mutation validation (Section 15.7) can be unified into a single coherent recoverability calculation.

15 Application: TCGA-KIRC

We now instantiate the framework on three real radiogenomic cohorts — TCGA-KIRC (clear-cell renal cell carcinoma) [42], a non–small-cell lung cancer (NSCLC) cohort [43], and ADNI (Alzheimer's Disease Neuroimaging Initiative) [44], spanning two tumour cohorts and a non-oncologic brain cohort — to demonstrate how the recoverability spectrum, the cross-validated and permutation-calibrated diagnostics of Section 12.6, and the dimensional-collapse warnings of Section 12.5 interact on typical cohorts. The deliverable is not a predictor but a calibrated answer to "how many genomic directions does imaging see, and how well." Two choices matter for getting a defensible answer. First, the imaging representation: a high-dimensional deep embedding ($d_{\mathrm{native}} = 1841$) places the fit deep in the saturating regime of Section 12.5, whereas a compact panel of organ-level radiomic descriptors ($d_{\mathrm{native}} = 107$) keeps it in the merely high-dimensional one. Second, the marginal transform: the Gaussian-copula normalization of Section 12.1 is essential because raw radiomic marginals have excess kurtosis in the tens. With both, the imaging-to-genomics channel is narrow but not empty: one genomic direction is image-identifiable above the permutation null in every preprocessing variant we tried, with a small (order 0.1) but honest cross-validated recoverability that, unlike the other two cohorts, survives demographic and scanner deconfounding (Section 15.6). We stress at the outset that this is a modest, single-direction effect, borderline in the default configuration, and report it as such rather than as a broad demonstration that imaging recovers genomics. All numbers below are reproducibly generated by the companion notebook and saved to `notebooks/figures/tcga_kirc/gene_expression/organ_radiomics/stats.json` (and the NSCLC counterpart).

15.1 Data and processing pipeline

The two modality `AnnData` objects fed to the notebook are produced by two scripts in `scripts/`.

Imaging (`process_imaging_tcga_kirc.py` $\to$ `make_imaging_matrix.py`). For each TCGA-KIRC patient we obtain the venous-phase abdominal CT volume (`0502_VENOUS.nii`) and a manually curated tumour mask. The pipeline:

1. Pull the canonical TCIA volumes [45, 46] and the radiologist-drawn tumour segmentations; orient every volume into the canonical nibabel orientation.
2. Run TotalSegmentator [47] with the `kidney_left`/`kidney_right` target list to produce an organ mask; combine with the tumour mask to form (tumour, organ, tumour$\cup$organ) mask triples per patient. Small blobs are removed, holes filled, and a morphological closing applied to stabilise mask boundaries.

3. Extract patient-level imaging features. Our primary representation is a panel of $d_{\text{native}} = 107$ *organ-level* PyRadiomics descriptors [30, 31] computed within the kidney organ mask (shape, first-order intensity, and GLCM / GLRLM / GLSZM / GLDM / NGTDM texture families). We deliberately prefer this compact, interpretable panel over a high-dimensional pretrained *RadImageNet* [32] embedding ($d_{\text{native}} = 1841$, supported for ablation): as Section 12.5 predicts and Section 15.3 confirms, the deep embedding drives the fit into the saturating regime where the spectrum is uninformative.
4. Stack patient-level features into an `AnnData` object whose `obs` index is the TCGA `patient_id` and whose `var` index is the radiomic feature names. Output: `data/tcga_kirc/imaging/organ_radiomics.h5ad`.

Genomics (`make_genomics_matrix.py`). For each matched patient we pull the TCGA-KIRC RNA-seq counts (raw STAR counts from the GDC) and the somatic-mutation MAFs.

1. Counts are CPM-normalised, $\log_{1+}$-transformed, and reduced to the top 2000 highly variable genes (Seurat-style HVG selection in `scanpy.pp.highly_variable_genes` [48]); the resulting matrix is stored in the primary `AnnData`.
2. Somatic mutations are folded into a sparse binary gene-symbol matrix on the same patient index for downstream per-gene validation (Section 15.7).
3. Output: `data/tcga_kirc/genomics/gene_expression.h5ad`.

After intersecting patients across both `AnnData` objects we retain $n = 190$ patients with $p_{\text{native}} = 2000$ HVG transcripts and $d_{\text{native}} = 107$ organ radiomic features.

15.2 Working-dimension budget

Both modalities are passed through the Gaussian-copula marginal transform of Section 12.1 and PCA-reduced symmetrically to working dimensions $p^\star = d^\star = 38 = \lfloor n/5 \rfloor$ following the $p^\star + d^\star \leq n/\tau$ rule of Section 12.5 ($\tau = 5$). This places the analysis at $\kappa_G + \kappa_X = 76/190 \approx 0.4$ — the high-dimensional regime, not the saturating one, but well inside the territory where in-sample canonical correlations are heavily biased upward.

15.3 The recoverability spectrum and its null

Fitting regularised CCA (Ledoit–Wolf shrinkage on both sides) on the copula-transformed, organ-radiomic representation returns an in-sample leading correlation $\hat{\rho}_1 \approx 0.76$ ($\hat{R}_1 \approx 0.58$) and a total in-sample mutual information of 3.9 bits over 12 retained components — far from the near-unit saturation a deep 1841-dimensional embedding produces at the same n, confirming that the compact panel keeps the fit out of regime 3. The in-sample numbers are still optimistic: 5-fold cross-validation deflates the leading recoverability to $\hat{R}_1^{\text{cv}} \approx 0.07$ in the default preprocessing, and the in-sample share of imaging variance attributable to genomics drops from 0.27 to 0.011 under cross-validation (Fig. 12a).

The substantive question is whether *any* direction survives an honest, scale-matched test. It does, but weakly, and the magnitude is preprocessing-dependent — so we report the robustness sweep rather than a single configuration. Across a grid of defensible choices (expression layer $\in$ {raw counts, TPM} $\times$ HVG method $\in$ {dispersion, variance} $\times$ 3 seeds; 12 configurations), the leading-three permutation p-value is below 0.05 in 12/12 configurations ($p < 0.01$ throughout), the best cross-validated recoverability has median $\hat{R}^{\text{cv}} = 0.11$ (range 0.06–0.15), and the honest effective image-identifiable rank has median 1 (range 0–3; ≥ 1 in 11/12). In the single default configuration the leading in-sample permutation p-values are 0.10, 0.015, 0.005 for the top three directions. The reading is consistent: one genomic direction is robustly image-identifiable above chance, carrying a small but real cross-validated recoverability of order 0.1, with no evidence for more than one or two.

15.4 An empirical upper bound on $I(G; X)$

Because the channel is no longer saturating, the permutation null on $\hat{\rho}_1$ is well-behaved — the per-permutation $\hat{\rho}_1^{\text{null}}$ has mean 0.68 and a 95% quantile of 0.81, not the near-1 value a deep embedding produces. Aggregating the cumulative mutual information $\text{MI}^{\text{null}}(K) = -\frac{1}{2}\sum_{i \leq K} \log(1 - \hat{\rho}^2_{i,\text{null}})$ over the top $K = 3$ tested components and reading off its 95th percentile gives a one-sided upper confidence limit (Fig. 12b)

$$I(G; X)\,\big|_{\text{KIRC, top-3}} \;\leq\; 1.58 \text{ bits} \quad (\text{one-sided } 95\% \text{ UCL}), \tag{26}$$

an order of magnitude tighter than the 21-bit bound the saturating deep embedding admits. The observed cumulative MI is 1.59 bits, sitting at the upper edge of the null (mean 0.82 bits in nats $\to$ bits) — the same borderline-positive picture as the spectrum.

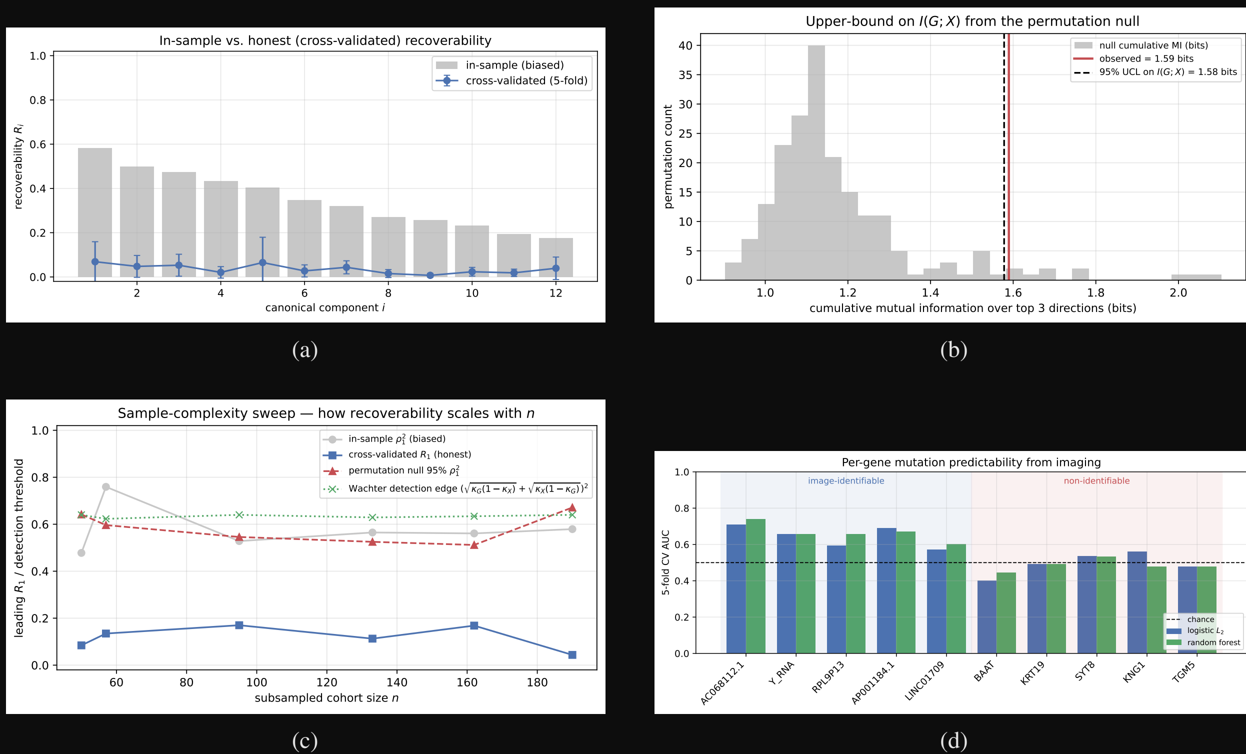


Figure 12: TCGA-KIRC results. **(a)** Recoverability spectrum (organ radiomics, copula marginals). In-sample $\hat{R}_i$ (grey) is biased upward by the high-dimensional geometry; cross-validated $\hat{R}_i$ (blue, 5-fold) is a small positive residual whose leading component is permutation-significant across preprocessing variants. The NSCLC leading direction (Fig. 15) is of similar magnitude but, unlike KIRC's, does not generalize or survive de-confounding at full n. **(b)** Cumulative-MI null distribution (top-3 canonical components). The observed value (red) sits at the upper edge of the null bulk; the dashed line marks the 95% upper confidence limit on $I(G;X)$ used in (26). **(c)** Sample-complexity sweep. Honest $\hat{R}_1^{\text{cv}}$ (blue squares) stays in a low band (≈ 0.04–0.17) across cohort sizes without a clear trend; in-sample $\hat{\rho}_1^2$ (grey) is dominated by the high-dimensional bulk; the permutation 95% null (red triangles) and Wachter edge (green crosses) sit above the honest recoverability at every n. **(d)** Per-gene predictability. Five high-recoverability and five low-recoverability genes, two classifiers, 5-fold CV. The high-recoverability bars clear the chance line at 0.5; the low-recoverability bars do not.

The *effective image-identifiable rank* of Section 12.6 — components whose cross-validated $\hat{R}_i^{\text{cv}}$ clears its own permutation null at $p < 0.05$ — has median $\hat{k}_{\text{eff}} = 1$ across the robustness grid (range 0–3). This is the empirical instantiation of the rank bound of Proposition 6 after sample-size deflation: one identifiable direction, not the dozen the native dimensions might suggest.

15.5 Sample-complexity sweep

A borderline single-n result is most informative when paired with the scaling in n. We subsample the cohort to $n \in \{50, 57, 95, 133, 162, 190\}$, refit the working pipeline (same $p^\star + d^\star \leq n/\tau$ budget, 5-fold CV, 100 permutations) and plot the observables vs. n (Fig. 12c). The cross-validated $\hat{R}_1^{\text{cv}}$ stays in the band ≈ 0.04–0.17 across the full range without a clear upward trend — because $p^\star, d^\star$ grow with n under our adaptive rule, more samples buy resolution at higher working dimension rather than a larger leading recoverability. The Wachter detection edge sits at ≈ 0.63 throughout, so distinguishing ρ_1 from the high-dimensional bulk *in-sample* would require $\rho_1^2 \gtrsim 0.6$; the honest signal lives well below that edge and is visible only through cross-validation and permutation, not the raw spectrum. Inverting the Wachter expression for a target $\rho_1^2 = 0.2$ at balanced $p^\star = d^\star$ gives the sample-size requirement $n \gtrsim 25\, p^\star$ quoted in Section 12.5: a larger cohort, or a smaller fixed working dimension, would be needed to lift this direction cleanly above the in-sample bulk.

15.6 Demographic and scanner deconfounding

The same skepticism we apply to NSCLC (scanner/batch, Section 16) and ADNI (age/sex, Section 17.4) must be applied here: a small channel is exactly the regime in which a demographic or acquisition confound could account for the whole effect. We therefore residualize *both* modalities (in the copula-transformed, PCA-reduced working space) on a design matrix of patient age, sex, ethnicity, and race — read from the GDC clinical table [42] — together with the per-patient scanner manufacturer (GE / Siemens / Philips) from the TCIA series digest [45, 46], all with full coverage on the $n = 190$ cohort, and refit the honest pipeline (1000 permutations).

Unlike the other two cohorts, the KIRC channel *survives*. The leading cross-validated recoverability is essentially unchanged — $\hat{R}_1^{\text{cv}} = 0.115 \to 0.148$ after deconfounding (cross-validated permutation p from 0.002 to 0.001) — and the effective image-identifiable rank does not drop. This is the decisive contrast of the paper: the controls that collapse NSCLC ($\hat{R}_1^{\text{cv}}$ non-significant after batch residualization) and ADNI ($0.37 \to 0.006$, $\hat{k}_{\text{eff}} : 2 \to 0$) leave the KIRC association intact, so the small KIRC channel is not explained by the obvious common causes (Fig. 13). We caveat appropriately: η^2 of the leading imaging PCs attributable to scanner is modest here, the cohort is single-disease, and unmeasured acquisition parameters (kVp, slice thickness, contrast timing) were not in the available headers, so "confound-robust" means robust to the measured demographics and scanner manufacturer, not to every conceivable batch effect.

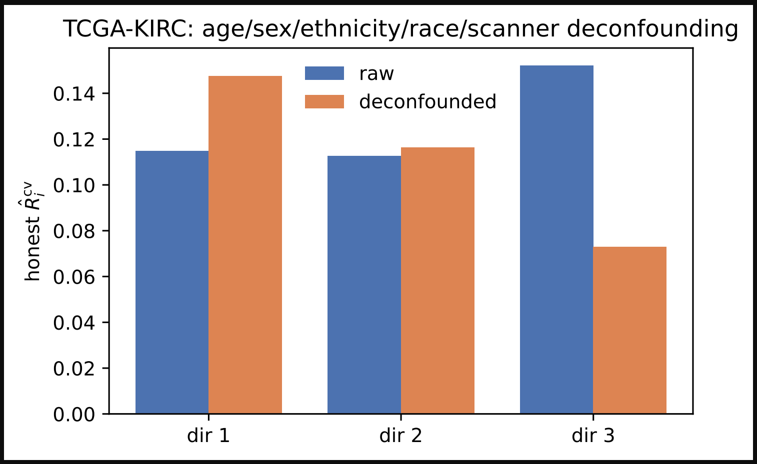


Figure 13: TCGA-KIRC demographic/scanner deconfounding. Honest cross-validated recoverability per direction, raw (blue) versus after regressing age, sex, ethnicity, race, and scanner manufacturer out of both modalities (orange). Unlike NSCLC (Section 16) and ADNI (Fig. 16d), the leading direction is undiminished ($\hat{R}_1^{\text{cv}} = 0.115 \to 0.148$, permutation $p \leq 0.002$ throughout): the small KIRC channel is not a demographic or scanner artefact.

15.7 Per-gene non-linear validation

The recoverability spectrum makes a sharp, falsifiable prediction: genes loading on *identifiable* directions should be predictable from imaging, genes on *non-identifiable* directions should not, and a flexible non-linear model should not substantially exceed the linear Bayes-optimal ceiling. We test this by ranking genes by their model recoverability $R(e_j)$, median-binarizing the five highest- and five lowest-ranked, and training L_2 logistic regression and a small random forest to predict the binarized expression from the imaging PCs (5-fold CV):

group	mean AUC (logistic)	mean AUC (RF)
image-identifiable	0.643	0.665
non-identifiable	0.493	0.485

The ordering is exactly as predicted: the identifiable group is decodable above chance (AUC ≈ 0.65) while the non-identifiable group sits at 0.5, and the random forest does not materially exceed the logistic ceiling — a direct, gene-level corroboration that the recoverability score, not gene identity per se, governs what imaging can read out (Fig. 12d).

15.8 Histopathology vs. radiology: the same channel through a microscope

The CT channel is narrow because a macroscopic attenuation map is a coarse view of a molecular state assayed in the tumour tissue itself. A natural test of this reading is to swap the imaging modality for digital *histopathology* — the H&E whole-slide images (WSI) of the same TCGA-KIRC tumours — while holding the genomic target, the preprocessing, and the working-dimension budget fixed. We pull the GDC diagnostic and tissue slides (580 "`.svs`" images over 190 cases, mapped to Case ID and tumour/normal status through the GDC `sample_sheet.tsv`), and extract an interpretable hand-crafted feature bank that mirrors the radiomics families — 54 descriptors in four groups: H&E *stain* (Ruifrok–Johnston colour-deconvolution optical densities), grayscale *first-order* intensity, *texture* (Haralick GLCM + rotation-invariant LBP), and nuclear *structure* (haematoxylin-dense connected-component morphology). A case's per-slide features are averaged over its tumour slides into one case-level vector (`scripts/build_kirc_wsi_features.py`, written to `data/tcga_kirc/wsi/wsi_tumor.h5ad`), so a WSI feature vector aligns to the same case's expression profile exactly as a radiomic vector does. We then run the identical honest pipeline (Gaussian-copula marginals, symmetric PCA to $p^\star = d^\star = \lfloor n/5 \rfloor$, 5-fold CV, 1000-permutation CV-null; `scripts/kirc_wsi_recoverability.py`).

The contrast is large and in the predicted direction. On the same expression target and the same n-budget, histopathology recovers the leading genomic direction roughly four times as strongly as CT: $\hat{R}_1^{\mathrm{cv}} = 0.46$ from WSI ($n = 189$) versus 0.11 from CT organ radiomics ($n = 190$), with the effective image-identifiable rank rising to $\hat{k}_{\mathrm{eff}} = 3$ (all three leading directions clear their CV-permutation null at $p \leq 0.003$) and the top-3 cumulative mutual information climbing from 1.6 to 2.2 bits (Fig. 14a). This is the same weak-but-real channel viewed at the resolution where the genomics is actually generated: the microscope sees the cellular morphology — nuclear density, stain distribution, texture — that co-varies with transcriptional state, where the CT voxel sees only its macroscopic shadow.

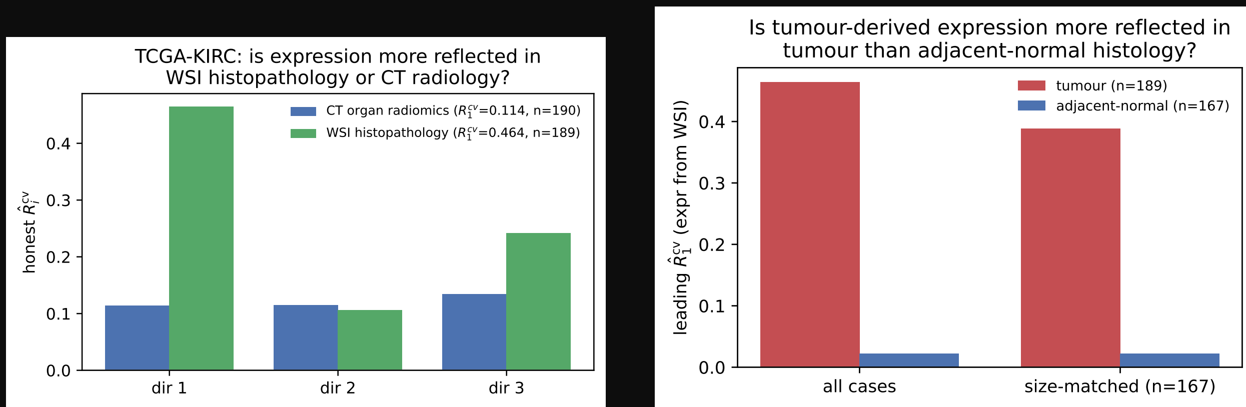
Tumour vs. adjacent-normal. If the histology–expression coupling is genuinely tumour-biological rather than a generic tissue or batch effect, a patient's tumour-derived expression should be reflected in their *tumour* slides but not in their *adjacent-normal* slides. It is: re-running the probe on each case's normal slides collapses the leading recoverability to the floor, $\hat{R}_1^{\mathrm{cv}} = 0.02$ ($n = 167$) against 0.46 for tumour — a gap that survives size-matching the larger group to the smaller (0.39 vs. 0.02 at $n = 167$, Fig. 14b), so it is not merely the larger tumour-slide count. The molecular signal is read out of the malignant tissue, not the patient's normal kidney.

Deconfounding. A four-fold-larger channel is also four times the opportunity for a confound, so we apply the same control that vindicated the CT channel (Section 15.6), with the WSI-appropriate batch term: age, sex, ethnicity, race, and — in place of CT scanner manufacturer — the TCGA *tissue source site* (the barcode institution code, a proxy for per-laboratory H&E staining and fixation batch). Residualizing both modalities on this design leaves the channel untouched: $\hat{R}_1^{\mathrm{cv}} = 0.45 \to 0.49$ (CV-permutation $p \leq 0.001$ throughout, $\hat{k}_{\mathrm{eff}}$ unchanged at 3). The histology–genomics coupling is therefore not an artefact of demographics or staining site; like the CT channel, it is robust to the measured common causes, but an order of magnitude wider.

15.9 Implications

Three concrete take-aways follow.

1. *A small but real channel.* For TCGA-KIRC at $n = 190$ with 2000 HVG-filtered transcripts and 107 organ radiomic features, the imaging-to-genomics channel admits a 95% upper bound $I(G; X) \leq 1.58$ bits over the leading three canonical directions (Eq. 26), with effective image-identifiable rank $\hat{k}_{\mathrm{eff}} = 1$ (median over preprocessing). The substantive claim is that exactly one genomic direction is robustly image-identifiable above chance (permutation $p < 0.05$ in 12/12 preprocessing variants), carrying a modest cross-validated recoverability $\hat{R}^{\mathrm{cv}} \approx 0.1$ — neither the dozen directions a 2000-gene panel might invite nor the flat null a saturating deep embedding produces. The signal survives dropping all shape features and residualizing every feature on organ volume, so it is texture/intensity, not size; and, decisively, it survives regressing out age, sex, ethnicity, race, and scanner manufacturer ($\hat{R}_1^{\mathrm{cv}} = 0.115 \to 0.148$, Section 15.6) — the controls that null both other cohorts. We nonetheless read it cautiously: the effect is small, of order $\hat{R}^{\mathrm{cv}} \approx 0.1$, borderline in the single default configuration (leading in-sample permutation $p = 0.10$), and rests for its robustness on the preprocessing grid rather than on any one fit.
2. *Design recommendation.* The sample-complexity sweep (Section 15.5) shows $\hat{R}_1^{\mathrm{cv}}$ in a low band (0.04–0.17) without a clear upward trend over $[50, 190]$; the Wachter edge analysis indicates that lifting $\rho_1^2 = 0.2$ cleanly above the in-sample bulk at a balanced working dimension $p^\star = d^\star$ requires $n \gtrsim 25\, p^\star$, an order of magnitude



(a) (b)

Figure 14: TCGA-KIRC histopathology channel. **(a)** Honest per-direction $\hat{R}_i^{\text{cv}}$ for the *same* HVG-expression target recovered from WSI histopathology (green) versus CT organ radiomics (blue), identical copula/PCA pipeline and n-budget; the leading histology recoverability is $\sim 4\times$ the radiology one. **(b)** The histology channel is tumour-specific: tumour slides (red) carry the expression signal that adjacent-normal slides (blue) do not, before and after size-matching. All numbers reproduced by `scripts/kirc_wsi_recoverability.py` (and `kirc_wsi_deconfound.py`), saved to `notebooks/figures/tcga_kirc/gene_expression/wsi/stats.json`.

beyond typical radiogenomic cohort sizes. The framework converts a borderline result into a quantitative cohort specification.

3. *Representation matters more than estimator.* The same cohort reads as a hard null under a 1841-dimensional deep embedding (saturating regime, $I(G;X) \leq 21$ bits, $\hat{k}_{\text{eff}} = 0$) and as a weak positive under a 107-feature organ radiomic panel with copula marginals — the difference is entirely the imaging representation and marginal transform, not the linear ceiling. Published per-gene AUCs of 0.65–0.80 at $n \sim 100$–300 [22, 23, 26–28] are consistent with the order-0.1 recoverabilities and AUC ≈ 0.65 we recover for the identifiable group (Section 15.7); claims *well* above that, at comparable cohort dimension, remain candidates for overfit or leakage.

16 Application: NSCLC

The non–small-cell lung cancer cohort [43] is the most instructive of the three: at small n it presents as the strongest positive, but the full cohort exposes it as a cohort-specific overfit that does not generalize — a cautionary counterpoint to the genuinely weak-but-robust KIRC channel, and a reminder that label-permutation significance alone is not proof of a real association. Imaging is the venous/contrast chest CT; we run TotalSegmentator [47] for the five pulmonary lobes, union them into a lung organ mask, resample to 2 mm isotropic (full-resolution shape extraction over the $\sim 10^7$-voxel lung is intractable), and extract the same $d_{\text{native}} = 107$ organ-level PyRadiomics panel. Genomics is the cohort RNA-seq, processed identically (CPM, $\log_{1+}$, 2000 HVGs). The full cohort with usable masks is $n = 129$ (one series, R01-052, dropped for a corrupted volume), with the copula transform and a $p^\star = d^\star = \lfloor n/5 \rfloor = 25$ working-dimension budget.

In-sample significance does not survive honest validation. Fitting returns an in-sample leading correlation $\hat{\rho}_1 = 0.73$ and 2.9 bits of in-sample MI; the patient-label permutation null is separated ($\hat{\rho}_1^{\text{null}}$ mean 0.66, 95% quantile 0.72), so the in-sample spectrum is nominally significant (top three $p = 0.030, 0.015, 0.030$). But this is the optimistic estimator, and the honest checks contradict it. The leading cross-validated recoverability is only $\hat{R}_1^{\text{cv}} = 0.11$, and — decisively — the held-out R^2 of the leading canonical genomic score is *negative* (-0.46): the direction that maximizes in-sample correlation predicts held-out imaging *worse* than a constant mean. The scale-matched cross-validated permutation test still flags the leading component ($p = 0.02$; the second and third are null, $p = 0.13, 0.97$), so $\hat{k}_{\text{eff}} = 1$ by that criterion, and the robustness grid concurs (median leading $\hat{R}^{\text{cv}} = 0.11$, range 0.10–0.16; rank ≥ 1 in 6/6 configs). The lesson is that label-permutation significance is necessary but not sufficient: a marginal $\hat{R}_1^{\text{cv}} \approx 0.11$ can clear its own permutation null while failing held-out prediction outright. The MI upper bound is correspondingly small,

$I(G;X)\big|_{\text{top-3}} \leq 1.23$ bits. The per-gene validation does show the identifiable group is internally consistent (mean AUC 0.70 logistic / 0.69 RF versus $0.51/0.51$), but as below this reflects the overfit axis rather than generalizable biology.

The signal is cohort-specific overfit, not biology. The negative held-out R^2 is explained by cohort heterogeneity. The cohort was released in two tranches; fit separately, the original 98 patients give $\hat{R}_1^{\text{cv}} \approx 0.25$–$0.29$ and the 31 later cases give 0.35, but *pooled* they give only 0.11 — lower than either part. Their leading genomic directions are near-orthogonal ($|\cos| = 0.07$): each tranche fits a strong but mutually inconsistent axis, the signature of cohort-specific overfit rather than a shared biological channel. The earlier small-n analysis of 90 patients — dominated by the more homogeneous original tranche — reported a clean $\hat{R}_1^{\text{cv}} \approx 0.33$ with held-out $R^2 = 0.40$; that estimate did not survive enlarging the cohort, which is exactly the kind of non-replication the honest diagnostics are meant to catch.

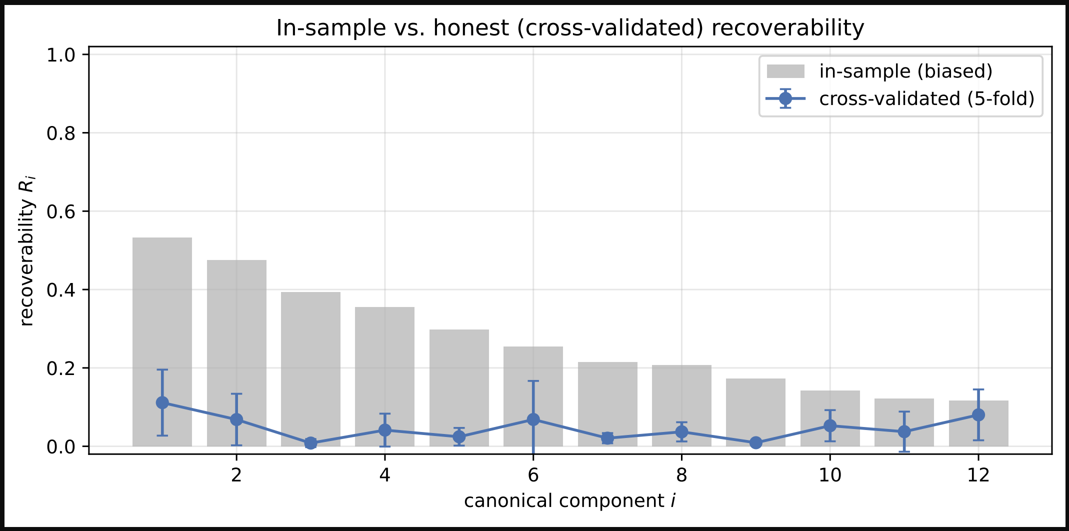


Figure 15: Recoverability spectrum on NSCLC ($n = 129$, organ radiomics, copula marginals). The leading cross-validated $\hat{R}_1^{\text{cv}} \approx 0.11$ (blue) is only marginally above the permutation null and, unlike at small n, does not generalize: the held-out R^2 of the leading direction is negative and the channel fails de-confounding.

De-confounding controls confirm the failure. Consistent with overfit, the leading direction also fails the scanner/batch controls that a genuine channel should survive. Scanner manufacturer (GE / Siemens / Philips) and convolution kernel are associated with both modalities (kernel explains $\eta^2 \approx 0.2$–0.3 of the leading principal components), so a naive estimate is partly batch-inflated. None of three de-confounding controls reaches significance. (i) Regressing manufacturer and kernel out of *both* modalities drops the leading recoverability to $\hat{R}_1^{\text{cv}} = 0.089$ ($p = 0.070$). (ii) A batch-stratified permutation null, in which genomic labels are permuted only *within* manufacturer$\times$kernel blocks so that any batch-mediated covariance is preserved under the null, gives null mean 0.054 (95% quantile 0.136) against the observed 0.111 ($p = 0.100$). (iii) The largest homogeneous stratum (GE scanners with the `STANDARD` kernel, $n = 80$) yields $\hat{R}_1^{\text{cv}} = 0.138$. The measurable batch covariates thus account for only part of the inflation — residualizing them leaves a non-significant remnant — while the larger driver, the near-orthogonal per-tranche axes above, points to unmeasured acquisition differences (tube voltage, dose, slice thickness, and contrast phase were unavailable in the headers) compounded by small-n, high-dimensional overfitting. The full-cohort verdict is therefore that NSCLC carries *no* de-confounding-robust, generalizable radiogenomic channel — the opposite of the small-n reading.

17 Application: ADNI

The two tumour cohorts pair a localized lesion with a matched tissue genomic assay. ADNI is a deliberately different test: structural brain MRI [44] against *whole-blood* Affymetrix microarray expression, a non-oncologic cohort where any imaging–genomics channel must run through systemic rather than local biology. It is also by far the largest cohort we analyze ($n = 702$), which lets the sample-complexity sweep show a clean scaling law rather than a flat band.

Imaging (`run_fastsurfer.py` $\rightarrow$ `build_adni_fastsurfer_features.py`). Each T1 volume is processed with FastSurfer [49], the deep-learning FreeSurfer surrogate, yielding a DKT-atlas segmentation (`aparc.DKTatlas+aseg`) and the bias-field-corrected conformed volume (`orig_nu.mgz`). We build a $d_{\text{native}} = 1022$ imaging panel per subject from two families: (i) *morphometry* — volume and intensity summaries for every segmented region from the standard `aseg+DKT.stats` table (380 features); and (ii) *texture/shape* — the full PyRadiomics panel (shape, first-order, GLCM/GLRLM/GLSZM/GLDM/NGTDM; 31) extracted from `orig_nu.mgz` within each of six bilateral regions most affected in Alzheimer's disease [50]: hippocampus, entorhinal cortex, fusiform gyrus, inferior temporal gyrus, precuneus, and posterior cingulate (642 features). Stacking gives `data/adni/imaging/fastsurfer.h5ad`, keyed by ADNI subject ID.

Genomics (`make_genomics_matrix.py`). ADNI gene expression is Affymetrix Human Gene microarray, already normalized to log-scale intensities and indexed by gene symbol (20,092 genes, collapsed from the original probes); we therefore disable the count-oriented CPM and second $\log_{1+}$ steps and reduce to the top 2000 HVGs before the Gaussian-copula transform. Symbol indexing also makes the leading genomic directions interpretable (Section 17.4). Intersecting on subject ID retains $n = 702$ subjects with $p_{\text{native}} = 2000$ transcripts and $d_{\text{native}} = 1022$ imaging features, PCA-reduced symmetrically to $p^\star = d^\star = \lfloor n/5 \rfloor = 140$ ($\kappa_G + \kappa_X = 280/702 \approx 0.40$, the same high-dimensional regime as the tumour cohorts).

17.1 The recoverability spectrum and its null

Fitting copula-regularized CCA returns an in-sample leading correlation $\hat\rho_1 = 0.87$ ($\hat R_1 = 0.76$) and 7.1 bits of in-sample MI over 12 components. The honest *raw* spectrum is the strongest of the three cohorts — though Section 17.4 will show it is almost entirely a demographic artefact: 5-fold cross-validation gives a leading recoverability $\hat R_1^{\text{cv}} = 0.37$ and a clearly resolved second direction $\hat R_2^{\text{cv}} = 0.17$, with the remainder collapsing to ≈ 0 (Fig. 16a). The permutation null is well-separated ($\hat\rho_1^{\text{null}}$ mean 0.76, 95% quantile 0.77) and the top three observed correlations all clear it at $p \approx 0.005$. The *effective image-identifiable rank*, defined by cross-validated $\hat R_i^{\text{cv}}$ exceeding its own per-component permutation null, is $\hat k_{\text{eff}} = 2$ — the only cohort here where imaging resolves more than a single genomic direction. The result is stable across the preprocessing grid (HVG method $\times$ 3 seeds, 6 configurations): all permutation $p < 0.05$, leading $\hat R^{\text{cv}}$ median 0.31 (range 0.28–0.36), and $\hat k_{\text{eff}}$ consistently 2.

17.2 Information budget and sample-complexity scaling

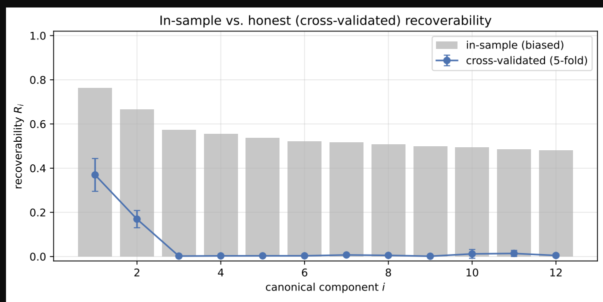
The observed cumulative mutual information over the top three components is 2.44 bits, sitting *above* the permutation band (null mean 1.69 bits, 95% upper limit 1.83 bits): unlike KIRC, where the observed value merely touched the upper edge of the null, here the channel is detected rather than bounded. Because ADNI is large, subsampling to $n \in \{50, 211, 351, 491, 597, 702\}$ traces a genuine scaling law rather than the flat band seen in KIRC (Fig. 16b): the honest $\hat R_1^{\text{cv}}$ climbs monotonically from 0.13 at $n = 50$ to a plateau of ≈ 0.35 by $n \approx 500$, while the in-sample $\hat\rho_1^2$ stays pinned near the Wachter detection edge (≈ 0.64) throughout. The leading direction lives *below* that in-sample edge at every n and is visible only through cross-validation and permutation — the same lesson as the tumour cohorts, but here the honest signal is large enough that even a few hundred subjects suffice to register it cleanly.

17.3 Per-gene non-linear validation

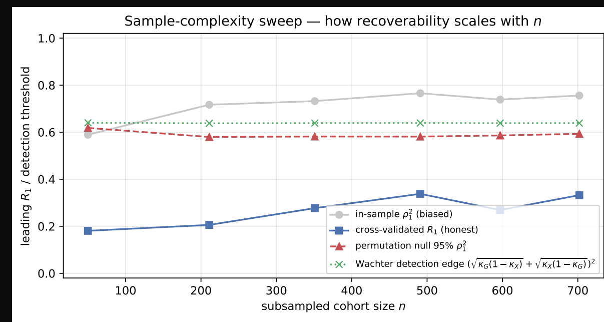
The recoverability ordering predicts gene-level decodability. Ranking transcripts by model recoverability $R(e_j)$, median-binarizing the five highest- and five lowest-ranked, and training L_2-logistic and random-forest classifiers to predict expression status from the imaging PCs (5-fold CV) gives a clean separation, sharper than in either tumour cohort:

group	mean AUC (logistic)	mean AUC (RF)
image-identifiable	0.842	0.841
non-identifiable	0.466	0.463

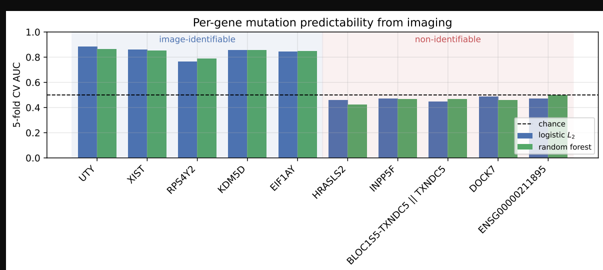
The image-identifiable transcripts are decodable from brain morphometry at AUC ≈ 0.84 while the non-identifiable controls sit at chance, and the random forest does not exceed the linear ceiling — exactly the falsifiable pattern the recoverability score predicts (Fig. 16c). This confirms the *internal consistency* of the recoverability ordering, but not its biological meaning: as Section 17.4 shows, the decodable transcripts are the ones correlated with age and sex, which brain morphometry reads out directly, so the high AUC is the per-gene shadow of the same demographic channel rather than an independent molecular signal.



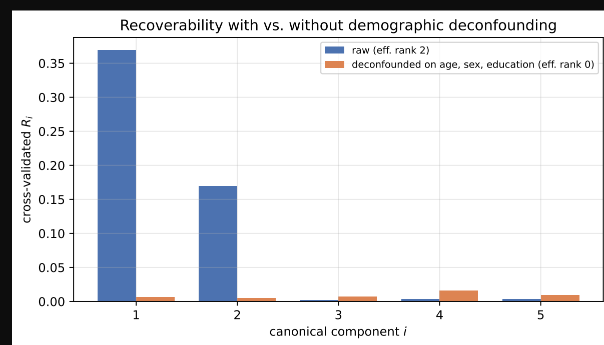
(a)



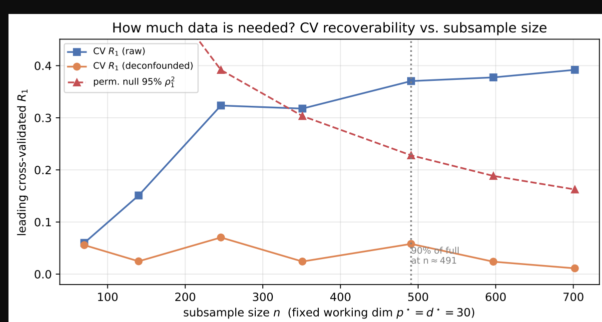
(b)



(c)



(d)



(e)

Figure 16: ADNI results ($n = 702$, FastSurfer morphometry + PyRadiomics, copula marginals). **(a)** Recoverability spectrum. The honest cross-validated spectrum (blue) resolves *two* genomic directions ($\hat R_1^{\mathrm{cv}} = 0.37$, $\hat R_2^{\mathrm{cv}} = 0.17$) above the permutation null before collapsing — the strongest and highest-rank channel of the three cohorts (compare Figs. 12a, 15). **(b)** Sample-complexity sweep. The honest $\hat R_1^{\mathrm{cv}}$ (blue squares) rises monotonically with cohort size to a plateau near 0.35, in contrast to the trendless KIRC band (Fig. 12c); the in-sample $\hat\rho_1^2$ (grey) hugs the Wachter edge (green) and the permutation 95% null (red) throughout. **(c)** Per-gene predictability. Five high- and five low-recoverability transcripts, two classifiers, 5-fold CV. The high-recoverability group reaches AUC ≈ 0.84; the low-recoverability group sits at the 0.5 chance line. **(d)** Demographic deconfounding. Honest cross-validated recoverability spectrum before (blue) and after (orange) regressing age, sex, and education out of both modalities. The raw rank-two channel ($\hat R_1^{\mathrm{cv}} = 0.37$) collapses to a null ($\hat R_1^{\mathrm{cv}} = 0.006$, $\hat k_{\mathrm{eff}} = 0$, permutation $p = 0.48$): the ADNI association is almost entirely demographically mediated. **(e)** Data-requirement sweep at fixed working dimension $p^\star = d^\star = 30$. The raw $\hat R_1^{\mathrm{cv}}$ (blue) clears the shrinking permutation null (red) near $n \approx 350$ and plateaus; the demographic-deconfounded $\hat R_1^{\mathrm{cv}}$ (orange) stays flat near zero at every n. More data detects the raw channel but never rescues the deconfounded one.

17.4 Demographic deconfounding: the channel is an age/sex artefact

A blood-to-brain channel of this strength warrants the same skepticism we applied to the NSCLC scanner confound, and here the verdict is sharper. Age, sex, and education are powerful common causes: they drive regional brain atrophy (hippocampal and entorhinal volume in particular, exactly where the leading imaging loadings fall) *and* systemic whole-blood expression. ADNI supplies these covariates [44], so we residualize age (year of expression collection minus birth year), sex, and years of education out of *both* modalities and refit the honest pipeline. The channel essentially vanishes (Fig. 16d): the leading cross-validated recoverability drops from $\hat{R}_1^{\mathrm{cv}} = 0.37$ to $\hat{R}_1^{\mathrm{cv}} = 0.006$ — only 1.7% of the raw value retained — the second direction collapses likewise, the cross-validated permutation p-value rises from 0.005 to 0.48, and the effective image-identifiable rank falls from $\hat{k}_{\mathrm{eff}} = 2$ to $\hat{k}_{\mathrm{eff}} = 0$. The gene-symbol indexing makes the mechanism explicit: the top loadings of the leading genomic direction are almost all sex-chromosome transcripts — the X-inactivation genes *XIST* and *TSIX* against the Y-linked *RPS4Y1*, *RPS4Y2*, *DDX3Y*, *UTY*, *EIF1AY*, *KDM5D*, *USP9Y*, *ZFY*, and *TXLNG2P* — so the leading "image-identifiable" axis is, transparently, sex, which brain morphometry reads out directly. The second direction is dominated by myeloid/inflammatory transcripts (*VSIG4*, *RNASE1*, *CCL3L/CCL4L*, *LGALS2*), consistent with an age-associated systemic-inflammation axis. Both are exactly the common causes the deconfounding removes. NSCLC fails its own de-confounding and held-out checks at full n (Section 16), and ADNI fails here even more sharply: the ADNI radiogenomic association does not survive deconfounding at all — what looked like the strongest channel of the three cohorts is, to within cross-validated noise, a demographic artefact. The model's value here is precisely that it converts an eye-catching $\hat{R}_1^{\mathrm{cv}} = 0.37$ into a calibrated null once the obvious common causes are removed.

17.5 How much data is needed?

Because ADNI is large, we can ask operationally how much of it a study like this actually requires. Holding the working dimension *fixed* at $p^\star = d^\star = 30$ (so that, unlike the sample-complexity sweep of Section 17.2, the only thing varying is n), we subsample the cohort over $n \in [70, 702]$ and track the honest leading recoverability, raw and demographic-deconfounded, against the permutation 95% null (Fig. 16e). Three things are visible. (i) The raw $\hat{R}_1^{\mathrm{cv}}$ rises steeply and first clears the permutation null near $n \approx 350$, reaches 90% of its full-cohort value by $n \approx 490$, and plateaus at ≈ 0.39 thereafter — so a few hundred matched subjects suffice to *detect* the raw channel, and more buy little. (ii) The permutation null falls monotonically with n (more samples, tighter null), which is what makes the fixed raw signal register as significant only past a few hundred subjects. (iii) The deconfounded curve sits flat near zero at *every* sample size: no amount of additional ADNI data would resurrect the channel, because the quantity being estimated is genuinely null. The practical reading is the mirror of the tumour cohorts: the binding constraint for ADNI is not sample size but confounding.

The contrast across cohorts is the useful object. Three cohorts, the same model, estimator, and working-dimension rule yield three distinct verdicts. In clear-cell RCC, a small but preprocessing-invariant single direction ($\hat{R}^{\mathrm{cv}} \approx 0.11$, $\hat{k}_{\mathrm{eff}} = 1$, perm $p \approx 0.005$) that survives demographic and scanner deconfounding (Section 15.6) — the most defensible positive of the three, though still a modest one. In NSCLC, an apparent signal that at small n looked the strongest ($\hat{R}_1^{\mathrm{cv}} \approx 0.33$, held-out $R^2 = 0.40$) but at the full $n = 129$ collapses to a marginal ≈ 0.11 with *negative* held-out R^2, near-orthogonal leading axes across release tranches ($|\cos| = 0.07$), and failed de-confounding ($p \geq 0.07$): cohort-specific overfit, not a shared channel. And an ADNI channel that is the strongest of all on raw data ($\hat{R}_1^{\mathrm{cv}} \approx 0.37$, $\hat{k}_{\mathrm{eff}} = 2$) yet collapses to a null once age, sex, and education are removed ($\hat{R}_1^{\mathrm{cv}} \approx 0.006$, $\hat{k}_{\mathrm{eff}} = 0$). The recoverable dimension is a property of the biology–imaging channel (Proposition 6), and the same honest pipeline both quantifies a small real low-rank association (KIRC) and exposes the two failure modes — non-replicating overfit (NSCLC) and confounding (ADNI) — that a naive in-sample estimator would have reported as strong positives. That discriminating power is exactly what the model is built to provide.

18 Conclusion

We reframed radiogenomics around an estimable, low-dimensional object — the recoverability spectrum $\{\rho_i^2\}$ and the image-identifiable genomic subspace — using a linear–Gaussian model whose closed forms reduce the whole analysis to canonical correlation analysis between the modalities. The mathematics is classical (Section 1); the contributions are the recoverability framing and rank limit, the scale-matched cross-validated / permutation pipeline that keeps high-dimensional inflation from masquerading as signal, and a three-cohort study that puts the pipeline to work.

The empirical message is deliberately conservative. Across TCGA-KIRC, NSCLC, and ADNI — analyzed with one model, estimator, and dimension-budget rule — only TCGA-KIRC yields a genomic direction that is simultaneously cross-validated, permutation-significant, and robust to demographic and scanner deconfounding, and even there the recoverability is small ($\hat{R}^{\text{cv}} \approx 0.1$, a single identifiable direction). The NSCLC "signal" is cohort-specific overfit (negative held-out R^2, near-orthogonal per-tranche axes) and the ADNI "signal" is an age/sex artefact that collapses under deconfounding. A naive in-sample analysis would have reported all three as strong, multi-direction positives. The framework's value is precisely this discrimination, and the broader take-away for the field is that confound-robust radiogenomic information is scarce, low-rank, and easily counterfeited at typical cohort sizes.

Limitations. Several caveats bound these conclusions. (i) The recoverability magnitudes are exact only under the linear–Gaussian law; the copula secures the marginals but the joint remains mildly heavy-tailed (Section 12.7), so reported MI values are approximate (the misspecification is signed conservatively, Eq. (21), but is nonzero). (ii) The conclusions depend on analyst choices — the working-dimension rule $p^\star = d^\star = \lfloor n/\tau \rfloor$, the imaging representation, the marginal transform — and while we report a robustness grid, the single KIRC positive is borderline in the default configuration. (iii) The per-gene validation ranks genes by an in-sample recoverability score; although evaluation is cross-validated, the selection is not fully nested, so it is best read as an internal-consistency check rather than an independent confirmation. (iv) "Confound-robust" means robust to the *measured* confounders; unmeasured acquisition parameters were unavailable for the tumour cohorts. (v) Two cohorts are small ($n = 190$, $n = 129$), at which the honest spectrum can certify only one or two directions even when more exist. These limitations all point the same way — toward larger, acquisition-harmonized cohorts with recorded covariates — which the sample-complexity analysis (Sections 15.5, 17.5) quantifies as the binding constraint.

Data Availability TCGA-KIRC imaging volumes are available from TCIA (`https://www.cancerimagingarchive.net/collection/tcga-kirc/`) [45, 46] and genomic data from the GDC portal (`https://portal.gdc.cancer.gov/`) [42]. The NSCLC radiogenomics cohort (imaging and expression) is available from TCIA (`https://www.cancerimagingarchive.net/collection/nsclc-radiogenomics/`) and GEO (https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE28827) [43]. ADNI data are available at `https://adni.loni.usc.edu/` [44].

Code Availability All analysis code and Lean [51] proof scripts for the main mathematical propositions are available at `https://github.com/pachterlab/rgit`.

A Symbol reference

$G \in \mathbb{R}^p,\ X \in \mathbb{R}^d$	per-patient genomic / imaging feature vectors
$\mathbf{G} \in \mathbb{R}^{n\times p},\ \mathbf{X} \in \mathbb{R}^{n\times d}$	stacked data matrices (n patients)
$A \in \mathbb{R}^{d\times p}$	genomics-to-imaging transfer map
σ^2	isotropic imaging noise variance
$\Sigma_G, \Sigma_X, \Sigma_{GX}$	genomic / imaging / cross covariance
$\Sigma_{G\mid X}$	posterior covariance of G given X (recoverability floor)
$B = \Sigma_{GX}\Sigma_X^{-1}$	Bayes-optimal linear map $x \mapsto \mathbb{E}[g \mid x]$
$M = \sigma^{-1} A \Sigma_G^{1/2}$	whitened transfer operator; d_i its singular values
$\tilde{G} = \Sigma_G^{-1/2} G$	whitened genomic vector
$\rho_i,\ R_i = \rho_i^2$	i-th canonical correlation / recoverability
$a_i,\ b_i$	i-th canonical genomic / imaging direction
$R(v)$	recoverability score of a named genomic direction v
$I(G; X)$	mutual information between modalities
k	dimension of the image-identifiable genomic subspace

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